

APPENDIX E

PATENT ASSIGNMENT COVER SHEET

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 Stylesheet Version v1.2

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SUBMISSION TYPE:	NEW ASSIGNMENT
NATURE OF CONVEYANCE:	ASSIGNMENT
CONVEYING PARTY DATA	
Name	Execution Date
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VINCENT J. MCGAHAY	02/24/2017
ZHONG-XIANG HE	02/23/2017
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State/Country:	CAYMAN ISLANDS
Postal Code:	KY1-1104
PROPERTY NUMBERS Total: 1	
Property Type	Number
Application Number:	15443276
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NAME OF SUBMITTER:	ANDREW M. CALDERON
SIGNATURE:	/Andrew M. Calderon/
DATE SIGNED:	02/27/2017
This document serves as an Oath/Declaration (37 CFR 1.63).	
Total Attachments: 3	
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source=IBUR032-US-NP_FiledAssignDec_02-27-17#page2.tif	
source=IBUR032-US-NP_FiledAssignDec_02-27-17#page3.tif	

**COMBINED DECLARATION and ASSIGNMENT (37 CFR 1.63) FOR UTILITY OR DESIGN APPLICATION
USING AN APPLICATION DATA SHEET (37 CFR 1.76)****Title of
Invention**

SEGMENTED GUARD-RING AND CHIP EDGE SEALS

DECLARATION

As the below named inventor, I hereby declare that:

This declaration
is directed to:

The attached application, or

United States application or PCT international application number _____ filed on

The above-identified application was made or authorized to be made by me.

I believe that I am the original inventor or an original joint inventor of a claimed invention in the application.

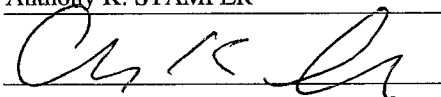
I hereby acknowledge that any willful false statement made in this declaration is punishable under 18 U.S.C. 1001 by fine or imprisonment of not more than five (5) years, or both.

I have reviewed and understand the contents of the application and am aware of the duty to disclose to the US Patent and Trademark Office all information known to me to be material to patentability as defined in 37 CFR § 1.56.

ASSIGNMENT

For good and valuable consideration, the receipt, sufficiency and adequacy of which are hereby acknowledged, I hereby sell, assign, transfer, to GLOBALFOUNDRIES Inc. having a business address of PO BOX 309, UGLAND HOUSE, GRAND CAYMAN, CAYMAN ISLANDS KY1-1104 its successors and assigns, absolutely and forever, my entire right, title and interest in and to the invention listed above, together with the Application, any and all Patents that may issue in the United States and/or any foreign countries, and any reissues, re-examinations, renewals, continuations, continuation-in-parts, divisionals or extensions thereof that may be issued or granted, and all right, title and interest to the inventions contained in said Patents and Applications, for the United States and all foreign countries, and all the rights and privileges relating thereto including but not limited to the priority rights arising from said Applications, the right to recover and take all such proceedings as may be necessary for the recovery of damages or otherwise in respect of past, present and future infringement, and the right to apply for, take and maintain patents on said inventions.

If any portion of this Combined Declaration or Assignment is held unenforceable, then remaining portions hereof shall nevertheless remain in full force and effect.

LEGAL NAME OF INVENTORInventor: Anthony K. STAMPERDate: Feb. 23, 2017Signature: 

Note: An application data sheet (PTO/SB/14 or equivalent), including naming the entire inventive entity, must accompany this form. Use an additional form for each additional inventor.

**COMBINED DECLARATION and ASSIGNMENT (37 CFR 1.63) FOR UTILITY OR DESIGN APPLICATION
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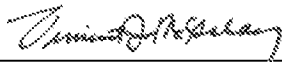
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LEGAL NAME OF INVENTOR

Inventor: Vincent J. MCGAHAY

Date : February 24, 2017

Signature: _____



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**COMBINED DECLARATION and ASSIGNMENT (37 CFR 1.63) FOR UTILITY OR DESIGN APPLICATION
USING AN APPLICATION DATA SHEET (37 CFR 1.76)****Title of
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SEGMENTED GUARD-RING AND CHIP EDGE SEALS

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The above-identified application was made or authorized to be made by me.

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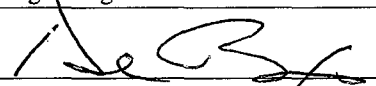
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LEGAL NAME OF INVENTORInventor: Zhong-Xiang HEDate : 02/23/2017Signature: 

Note: An application data sheet (PTO/SB/14 or equivalent), including naming the entire inventive entity, must accompany this form. Use an additional form for each additional inventor.

Doc Code: PA..

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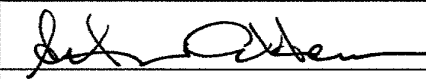
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Application Number	Not Yet Assigned
Filing Date	Concurrently Herewith
First Named Inventor	Anthony K. STAMPER et al.
Title	SEGMENTED GUARD-RING AND CHIP EDGE SEALS
Art Unit	Not Yet Assigned
Examiner Name	Not Yet Assigned
Attorney Docket Number	IBUR032-US-NP

SIGNATURE of Applicant or Patent Practitioner

Signature		Date (Optional)	2/27/2017
Name	Andrew M. Calderon	Registration Number	38,093
Title (if Applicant is a juristic entity)			

Applicant Name (if Applicant is a juristic entity)

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*Total of 1 forms are submitted.

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I hereby revoke all previous powers of attorney given in the application identified in either the attached transmittal letter or the boxes below.

Application Number	Filing Date

(Note: The boxes above may be left blank if information is provided on form PTO/AIA/82A.)

☒ I hereby appoint the Patent Practitioner(s) associated with the following Customer Number as my/our attorney(s) or agent(s), and to transact all business in the United States Patent and Trademark Office connected therewith for the application referenced in the attached transmittal letter (form PTO/AIA/82A) or identified above:

OR

133809

☐ I hereby appoint Practitioner(s) named in the attached list (form PTO/AIA/82C) as my/our attorney(s) or agent(s), and to transact all business in the United States Patent and Trademark Office connected therewith for the patent application referenced in the attached transmittal letter (form PTO/AIA/82A) or identified above. (Note: Complete form PTO/AIA/82C.)

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I am the Applicant (if the Applicant is a juristic entity, list the Applicant name in the box):

GLOBALFOUNDRIES Inc.

- ☐ Inventor or Joint Inventor (title not required below)
- ☐ Legal Representative of a Deceased or Legally Incapacitated Inventor (title not required below)
- ☒ Assignee or Person to Whom the Inventor is Under an Obligation to Assign (provide signer's title if applicant is a juristic entity)
- ☐ Person Who Otherwise Shows Sufficient Proprietary Interest (e.g., a petition under 37 CFR 1.46(b)(2) was granted in the application or is concurrently being filed with this document) (provide signer's title if applicant is a juristic entity)

SIGNATURE of Applicant for Patent

The undersigned (whose title is supplied below) is authorized to act on behalf of the applicant (e.g., where the applicant is a juristic entity).

Signature

Date (Optional)

Name

David Cain

Title

Senior Counsel, IP Law

NOTE: Signature - This form must be signed by the applicant in accordance with 37 CFR 1.33. See 37 CFR 1.4 for signature requirements and certifications. If more than one applicant, use multiple forms.

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This collection of information is required by 37 CFR 1.131, 1.32, and 1.33. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 3 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.

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ABSTRACT

The present disclosure relates to semiconductor structures and, more particularly, to segmented guard-ring and chip edge seals and methods of manufacture. The structure includes: a guard ring structure formed in a low-k dielectric material; and an edge seal structure formed through the low-k dielectric material to at least a substrate underneath the low-k dielectric material.

CLAIMS

What is claimed:

1. A structure comprising:
a guard ring structure formed in a low-k dielectric material; and
an edge seal structure formed through the low-k dielectric material to at least a substrate underneath the low-k dielectric material.
2. The structure of claim 1, wherein the low-k dielectric material is low-K SiCOH or p-SiCOH middle of the line and/or back end of the line dielectric material.
3. The structure of claim 1, wherein the substrate is a silicon on insulator substrate and the edge seal structure terminates below the low-k dielectric material.
4. The structure of claim 3, wherein the guard ring structure is a segmented guard ring structure.
5. The structure of claim 4, wherein the edge seal structure is a trench lined with a passivation layer and filled with a polyimide.
6. The structure of claim 5, wherein the polyimide is planarized above a last metal layer or bond pad.

7. The structure of claim 1, wherein the edge seal structure is a trench filled with oxide material which is also covering a metal insulator metal structure.
8. The structure of claim 7, wherein the trench filled with the oxide is below a last metal layer or bond pad.
9. The structure of claim 8, wherein the trench filled with the oxide includes an airgap.
10. The structure of claim 1, wherein the edge seal structure is one or more airgaps formed in the low-k dielectric material that is pinched off with oxide.
11. The structure of claim 10, wherein the oxide is a PECVD silane oxide.
12. A structure comprising:
 - a silicon on insulator substrate;
 - a middle of the line and back end of the line dielectric low-k dielectric material on the silicon on insulator substrate;
 - a wiring structure formed in the middle of the line and back end of the line dielectric low-k dielectric material to the silicon on insulator substrate;
 - a segmented guard ring structure formed in the middle of the line and back end of the line dielectric low-k dielectric material; and
 - an edge seal structure formed extending through the middle of the line and back end of the line dielectric low-k dielectric material.

13. The structure of claim 12, wherein the edge seal structure is a trench lined with a passivation layer and filled with a polyimide which has a planarized surface above a last metal layer or bond pad.

14. The structure of claim 13, wherein the edge seal structure terminates below the middle of the line and back end of the line dielectric low-k dielectric material

15. The structure of claim 12, wherein the edge seal structure is a trench filled with oxide material which covers a metal insulator metal structure below a last metal layer or bond pad.

16. The structure of claim 15, wherein the trench filled with the oxide includes an airgap.

17. The structure of claim 12, wherein the edge seal structure is one or more airgaps formed in the low-k dielectric material.

18. The structure of claim 17, wherein the one or more airgaps are at a lower level of the wiring structure.

19. The structure of claim 17, wherein the one or more airgaps are pinched off with PECVD silane oxide.

20. A method comprising:

forming a middle of the line and back end of the line dielectric low-k dielectric material on the silicon on insulator substrate;

forming a wiring structure formed in the middle of the line and back end of the line dielectric low-k dielectric material to the silicon on insulator substrate;

forming a segmented guard ring structure formed in the middle of the line and back end of the line dielectric low-k dielectric material; and

forming an edge seal structure extending through the middle of the line and back end of the line dielectric low-k dielectric material.

SEGMENTED GUARD-RING AND CHIP EDGE SEALS

FIELD OF THE INVENTION

[0001] The present disclosure relates to semiconductor structures and, more particularly, to segmented guard-ring and chip edge seals and methods of manufacture.

BACKGROUND

[0002] Guard ring technologies, e.g., segmented guard rings, are used in many different semiconductor technologies such as GaAs, SiGe, RFCMOS, and RFSOI. For example, radio frequency (RF) circuits require segmented guard rings to avoid parasitic coupling and inductive coupling (the latter phenomenon which can occur with continuous guard rings).

[0003] RF technologies typically comprise non-porous dielectric material at the middle of the line (MOL) and back end of the line (BEOL) processes. To make further improvements to RF technologies, e.g., improvements in RF circuitry performance, low-k dielectric, porous material at the MOL or BEOL processing levels is required. The low-k dielectric, porous material, e.g., SiCOH or p-SiCOH, is widely used in advanced CMOS technologies such as 90 nm and smaller generations for several BEOL wiring levels.

[0004] Particularly useful in these low-k dielectric, porous applications is a guard ring to prevent moisture ingress which can cause SiCOH cracking or degraded copper reliability (e.g., electromigration, stress migration, etc.). But the use of a segmented guard ring can cause reliability fails in these low-k dielectric, porous applications. Accordingly, RFSOI

chips with segmented guard ring and low-K SiCOH or p-SiCOH MOL/BEOL are known to be incompatible.

SUMMARY

[0005] In an aspect of the disclosure, a structure comprises: a guard ring structure formed in a low-k dielectric material; and an edge seal structure formed through the low-k dielectric material to at least a substrate underneath the low-k dielectric material.

[0006] In an aspect of the disclosure, a structure comprises: a silicon on insulator substrate; a middle of the line and back end of the line dielectric low-k dielectric material on the silicon on insulator substrate; a wiring structure formed in the middle of the line and back end of the line dielectric low-k dielectric material to the silicon on insulator substrate; a segmented guard ring structure formed in the middle of the line and back end of the line dielectric low-k dielectric material; and an edge seal structure formed extending through the middle of the line and back end of the line dielectric low-k dielectric material.

[0007] In an aspect of the disclosure, a method comprises: forming a middle of the line and back end of the line dielectric low-k dielectric material on the silicon on insulator substrate; forming a wiring structure formed in the middle of the line and back end of the line dielectric low-k dielectric material to the silicon on insulator substrate; forming a segmented guard ring structure formed in the middle of the line and back end of the line dielectric low-k dielectric material; and forming an edge seal structure extending through the middle of the line and back end of the line dielectric low-k dielectric material.

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] The present disclosure is described in the detailed description which follows, in reference to the noted plurality of drawings by way of non-limiting examples of exemplary embodiments of the present disclosure.

[0009] FIGS. 1A-1C show structures and respective fabrication processes in accordance with aspects of the present disclosure.

[0010] FIG. 2 shows an edge seal structure, amongst other features, and respective fabrication processes in accordance with aspects of the present disclosure.

[0011] FIG. 3 shows a structure and respective fabrication processes in accordance with additional aspects of the present disclosure.

[0012] FIG. 4 shows a structure and respective fabrication processes in accordance with additional aspects of the present disclosure.

[0013] FIGS. 5A-5D show structures and respective fabrication processes in accordance with additional aspects of the present disclosure.

DETAILED DESCRIPTION

[0014] The present disclosure relates to semiconductor structures and, more particularly, to segmented guard-ring and chip edge seals and methods of manufacture. More specifically, the present disclosure provides segmented guard-ring and chip edge seals with low-K dielectric material in middle of the line (MOL) and/or back end of the line

(BEOL) processing. Advantageously, the present disclosure provides compatibility of segmented guard ring with RFSOI chips comprising low-K SiCOH or p-SiCOH MOL/BEOL materials and fabrication processing.

[0015] In embodiments, the chip edge sealing structures can be provided in a final passivation nitride/polyimide film, which requires only one mask. This chip edge sealing structure can eliminate the front end of the line (FEOL) buried silicon on insulator (SOI) contact or edge seal, which is filled with a semiconductor or conductor, which connects the SOI top silicon with the silicon handle wafer through the buried oxide (BOX) contact in current SOI technologies, and can be used in advanced CMOS technologies (e.g., 14 nm) with non-segmented guard ring designs.

[0016] In further embodiments, a chip edge sealing structure can be provided post metal-insulator-metal (MIM) plate formation and prior to via interlevel dielectric (ILD) deposition. In this implementation, a thick ILD oxide deposition can be used to fill a trench to form the chip edge sealing. In yet another embodiment, the chip edge sealing structure can be provided with an airgap. In this implementation, a single airgap mask can be used, shared during low Coff SOI switch formation. In addition, in this implementation a thin (e.g., approximately 5 nm) layer of material, e.g., Al_2O_3 , can be provided for optimal moisture barrier protection. Advantageously, the processes and resultant structure provided herein can eliminate the use of an optional FEOL buried layer isolation edge seal discussed above. In addition, the edge seal structure can be used with both a segmented and non-segmented guard ring structures.

[0017] The guard-ring and chip edge seal structures of the present disclosure can be manufactured in a number of ways using a number of different tools. In general, though, the methodologies and tools are used to form structures with dimensions in the micrometer and nanometer scale. The methodologies, i.e., technologies, employed to manufacture the guard-ring and chip edge seal structures of the present disclosure have been adopted from integrated circuit (IC) technology. For example, the structures are built on wafers and are realized in films of material patterned by photolithographic processes on the top of a wafer. In particular, the fabrication of the guard-ring and chip edge seal structures uses three basic building blocks: (i) deposition of thin films of material on a substrate, (ii) applying a patterned mask on top of the films by photolithographic imaging, and (iii) etching the films selectively to the mask.

[0018] FIGS. 1A-1C show structures and respective fabrication processes in accordance with aspects of the present disclosure. FIG. 1A is a cross-sectional view of the structure and FIG. 1C is a top view of the structure shown in FIG. 1A. FIG. 1B is a partially exploded view of a guard ring structure and trench formed in a low-k dielectric material. As shown in FIG. 2, for example, the trench is filled with material chip edge seal.

[0019] In particular, the structure 10 shown in FIGS. 1A-1C includes a silicon-on-insulator (SOI) substrate 12. In embodiments, the SOI substrate 12 includes a handle wafer 14 and a buried insulator layer 16 under a semiconductor layer 18. In embodiments, the buried insulator layer 16 can be silicon dioxide or sapphire; however, it should be understood that the choice of insulator and handle wafer depends largely on the intended application. For example, for layer 16, sapphire is used for high-performance radio

frequency (RF) and radiation-sensitive applications; whereas, silicon dioxide is used for diminished short channel effects in microelectronics devices, as examples. In embodiments, handle wafer 14 can be silicon or sapphire. The semiconductor layer 18 may be composed of any suitable semiconductor material including, but not limited to, Si, SiGe, SiGeC, SiC, GaAs, InAs, InP, etc. The SOI substrate 12 can be formed by any conventional fabrication processes, e.g., wafer bonding, Separation by IMplantation of OXygen (SiMOX), etc.

[0020] Still referring to FIGS. 1A-1C, a low-k dielectric material 20 is formed over the semiconductor layer 18 in MOL and/or BEOL processes. In embodiments, the low-k dielectric material 20 can be a non-porous or porous polymer, carbon doped oxide, or oxide or combinations thereof, deposited using conventional deposition methods. In more specific embodiments, the low-k dielectric material 20 can be low-K SiCOH or porous-SiCOH materials. The low-k dielectric material 20 can be deposited by a chemical vapor deposition process.

[0021] A plurality of conductive wiring and via structures 22 (herein after referred to generally as wiring structure) and guard ring structure 24 are formed in the low-k dielectric material 20 to the semiconductor layer 18. The wires or vias could be formed either in low-k or oxide intermetal dielectrics with one or more of the wires or vias formed in low k dielectric. The guard ring structure 24 is preferably a segmented guard ring structure formed from metal vias and wiring layers, positioned near an edge of the chip. The guard ring structure 24 can be segmented to reduce rf coupling. More specifically, the guard ring structure 24 is a metal structure formed outside of the wiring

structure 22 near an edge of the chip using conventional damascene or dual damascene processes. The wiring structure 22 can also be formed by conventional damascene or dual damascene processes.

[0022] By way of example, the wiring structure 22 and segmented guard ring structure 24 can be formed using conventional deposition, lithography and etching processes (e.g., damascene or dual damascene processes). Specifically, after each layer of low-k dielectric material 20 is deposited, a resist deposited on a surface of the low-k dielectric layer 20 is exposed to energy (light) to form a pattern (opening). An etching process with a selective chemistry, e.g., reactive ion etching (RIE), will be used to form one or more vias and trenches in the low-k dielectric material 20 through the openings of the resist. In embodiments, the lithography and etching step can be a single or damascene process. The resist can then be removed by a conventional oxygen ashing process or other known stripants. Following the resist removal, conductive material, e.g., copper, aluminum, tungsten, etc., is deposited in the vias and trenches by conventional deposition processes, e.g., chemical vapor deposition (CVD) processes. Any residual metal material on the surface of the low-k dielectric material 20 can be removed by conventional chemical mechanical polishing (CMP) processes. This process continues until all of the wiring layers (e.g., M1 level, M2 level, etc.) are formed.

[0023] FIGS. 1A-1C further show a last metal layer or bond pad 25 formed in electrical and direct contact with at least the wiring structure 22 and the guard ring structure 24. The last metal layer or bond pad 25 can be formed by any subtractive method, as an example. Illustratively, in the implementation of the bond pad, an aluminum material can

be deposited on the surface of the low-k dielectric material 20, followed by a patterning and etching process (e.g., lithography and RIE process). An oxide and/or nitride layer 26 is formed over the last metal layer or bond pad 25 and any exposed surfaces of the low-k dielectric material 20. The oxide layer 26 can be deposited by a plasma enhanced (PE) CVD process to a thickness of about 0.5 microns; although other dimensions are also contemplated by the present disclosure.

[0024] Still referring to FIGS. 1A-1C, a trench 28 is formed through the layers of the structure 10 and preferably below the low-k dielectric material 20. In more specific embodiments, the trench 28 is formed through the low-k dielectric material 20 and extends to the semiconductor layer 18 and even more preferably to the buried insulator layer 16 or underneath the buried insulator layer 16. The trench 28 can have a width of about 2 microns; although other dimensions are contemplated herein. In embodiments, the trench 28 can be formed by conventional lithography and etching processes as should be understood by those of skill in the art.

[0025] As shown in FIG. 2, the trench 28 will be used to form an edge seal structure 30, at an outermost portion of the chip with respect to the wiring structure 22 and the guard ring structure 24. More specifically, the edge seal structure 30 is formed by deposition of materials within the trench 28. For example, the edge seal structure 30 includes a thin passivation layer 32, e.g., about 0.5 microns, formed on the sidewalls of the trench 28. In embodiments, the passivation layer 32 can be an oxide material or a nitride material or combinations thereof, deposited by a conventional PECVD process. The PECVD process will also deposit the passivation layer 32 over the oxide layer 26. Following the

deposition of the passivation layer 32, a polyimide film is deposited within the remaining portions of the trench 28, e.g., filling the trench 28. The polyimide or other polymer film 34 used for the final passivation on the wafer film 34 can be a spin on organic material (or other material). The films 32 and/or 34 should be impermeable to water to prevent water or moisture ingress through the low k material into the chip.

[0026] In this way, the processes and resultant structure provided herein eliminates the use of an optional front end of the line (FEOL) buried layer isolation edge seal. In addition, the edge seal structure 30 can be used with both a segmented and non-segmented guard ring structure 24.

[0027] FIG. 3 shows a structure and respective fabrication processes in accordance with additional aspects of the present disclosure. In the structure 10' of FIG. 3, the edge seal structure 30' is formed from an oxide or nitride material 36, after the fabrication of a metal insulator metal (MIM) capacitor structure 38 and several layers of the wiring structure 22 and the guard ring structure 24, e.g., M1 and M2 levels of the wiring structure 22 and the guard ring structure 24. The MIM capacitor structure 38 could be any BEOL passive device, such as a thin film resistor or a wire. As in the previous aspect, the wiring structure 22 and the guard ring structure 24 are formed to the semiconductor layer 18.

[0028] By way of more specific example of the embodiment of FIG. 3, after formation of the MIM or other passive structure 38, a trench 28' is formed through the layers of the structure 10' and preferably through and below the low-k dielectric material 20. In more specific embodiments, the trench 28' is formed through the low-k dielectric material 20

and extends to the semiconductor layer 18 and even more preferably to the buried insulator layer 16 or underneath the buried insulator layer 16. In embodiments, the trench 28' can be formed by conventional lithography and etching processes as should be understood by those of skill in the art. As in the previous embodiments, the trench 28' can have a width of about 2 microns; although other dimensions are contemplated herein.

[0029] After formation of the trench 28', oxide and/or nitride material 36 is deposited over the MIM structure 38 and within the trench 28' using a conventional CVD, physical vapor deposition, or similar process. In embodiments, the deposition of oxide may result in a pinch-off of the material 36 within the trench 28'. This pinch-off, in turn, will form an airgap 40 within the trench 28'. Accordingly, the airgap 40 has a smaller dimension than the filled trench 28'. The oxide material can undergo an optional CMP process, resulting in a planar surface for additional processing as noted herein. In this way, the edge seal structure 30' will be positioned at an outermost portion of the chip with respect to both the wiring structure 22 and the segmented guard ring structure 24.

[0030] Still referring to FIG. 3, the upper portions (e.g., the third or above metal wire/via levels) of the wiring structure 22 and the segmented guard ring structure 24 can be formed on the planarized oxide or nitride material 36 using conventional lithography, etching and deposition processes. The last metal layer or bond pad 25 is formed in electrical and direct contact with at least the wiring structure 22 and the guard ring structure 24. The last metal layer, shown as the third metal layer in FIG. 3, or bond pad 25 can be formed by any subtractive method as described herein. The dielectric layer 26 is formed over the last metal layer or bond pad 25 and any exposed surfaces of the low-k

dielectric material 20. The oxide layer 26 can be deposited by a PECVD process to a thickness of about 0.5 microns; although other dimensions are also contemplated by the present disclosure. A thin passivation layer 32, e.g., about 0.5 microns, is formed over the oxide layer 26 and a polyimide or other optional polymer film 34 is deposited on the passivation layer 32. The polyimide film 34 can be a spin on organic material (or other material).

[0031] FIG. 4 shows a structure and respective fabrication processes in accordance with additional aspects of the present disclosure. In the structure 10" of FIG. 4, an airgap or oxide lined airgap 30" can be formed in the low-k dielectric material 20. For example, after forming a first or multiple levels of the wiring structure 22 and the segmented guard ring structure 24 in the low-k dielectric material 20, e.g., SiCOH, using single or dual damascene processes, a trench is formed into the low-k dielectric material 20 to a segmented opening in the segmented guard ring 24. In embodiments, the trench is formed using conventional lithography and etching (RIE) processes as already described herein. The trench can be filled with an oxide material until it becomes pinched-off, thereby creating the airgap 30" or can be completely filled with oxide. For example, a PECVD silane oxide can be deposited within the trench to form the airgap 30". In this way, the airgap 30" can be formed at the M1 level (lower level) of the wiring structure 22. Two parallel airgaps 30'' are shown for example but any number of airgaps, e.g., one or more, may be employed and the top view layout of the airgaps could be rectangular (shown), circular, polygonal, etc.

[0032] After formation of the airgap 30", an oxide material can blanket deposited above the airgap 30" to form an interlevel dielectric layer 20'. In embodiments, the interlevel dielectric layer 20' is not a low-k dielectric material 20. The remaining wiring structures can be fabricated in the manner already described herein, including upper levels of the wiring structure 22 (M2 level and above) and the segmented guard ring structure 24, as well as the last metal layer or bond pads 25, etc. For example, by using lithography, etching and deposition steps, as described herein, the upper wiring structures, e.g., M2 level and above, can be fabricated in the dielectric layer 20'.

[0033] In alternative embodiments of FIG. 4, a dual damascene wiring structure 22 and segmented guard ring structure 24 is formed in the MOL in the low-k dielectric material 20, e.g., SiCOH, following by deposition of an NBLok 20', e.g., SiC_xNyHz, over the dual damascene wiring structure 22 and segmented guard ring structure 24. A trench is formed into the low-k dielectric material 20 (and NBLok 20') to a segmented opening in the segmented guard ring 24. In embodiments, the trench is about 0.3 μm wide; although other dimensions are contemplated herein. In embodiments, a trench can already be etched down to PC in NFET switches to form an airgap, e.g., about 0.18 μm deep trench. A PECVD silane oxide is then deposited to pinch off an airgap 30". In an addition implementation, a thin (e.g., approximately 5 nm) layer of material 42, e.g., Al₂O₃, can be provided in the semiconductor layer 18 and insulator layer 16 for perfect moisture barrier protection (at an outermost portion of the chip). The remaining structure 10" can be fabricated in the manner already described herein, including upper portions of the wiring structure 22 (e.g., M2 level and above) and the segmented guard ring structure 24 in the low-k dielectric material 20, as well as the last metal layer or bond pads 25, etc.

Alternatively, if the low k dielectric is used for multiple wiring level, then the airgap would extend through the same multiple wiring levels.

[0034] FIGS. 5A-5D show different views of structures and respective fabrication processes in accordance with additional aspects of the present disclosure. In addition, more details Specifically, FIG. 5A is top view of a chip edge seal and segmented guard ring structure. FIG. 5B is an exploded view of the structures shown in FIG. 5B, and FIGS. 5C and 5D are cross-sectional views along lines A-A and B-B, respectively, of FIG. 5B.

[0035] In particular, FIGS. 5A-5D show a structure 100 comprising an airgap 50 preferably placed over and touching substrate 98 or feature 105 on the substrate 98 such as a FET, resistor, capacitor, etc. formed on the silicon substrate and, more particularly, landing on a nitride film 95' over the substrate 98 or feature 100 (FIG. 5C). When specifying the airgap touching substrate 98 or feature 100, this means touching or etching into nitride layer 95' so that there is no low k dielectric 97 left under the airgap. The nitride layer 95' could be composed of any non-low-K dielectric, such as oxide, SiC, SiCN, etc. For the option of landing the trench etch on the feature 100, the trench etch, e.g., RIE, used in the fabrication process of the airgap 50 can be shallower, hence saving time and manufacturing costs. In embodiments, the airgap 50 can also land on other locations of the structure or can be etched into the substrate 98.

[0036] The airgap 50, as with the other aspects of the present disclosure forming the edge seal structure, is provided in a low-k dielectric material 20, e.g., SiCOH (FIGS. 5A-5D). The airgap 50 can be lined with an oxide material 110, which also is used for, e.g., an upper wiring level (e.g., M2 wiring layer) over the low-k dielectric material 20 (FIGS.

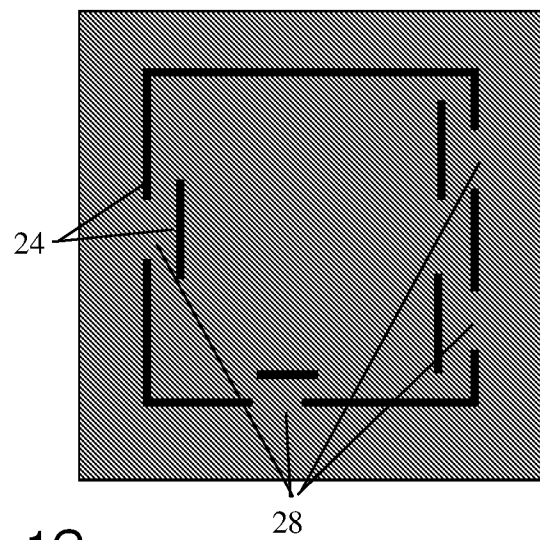
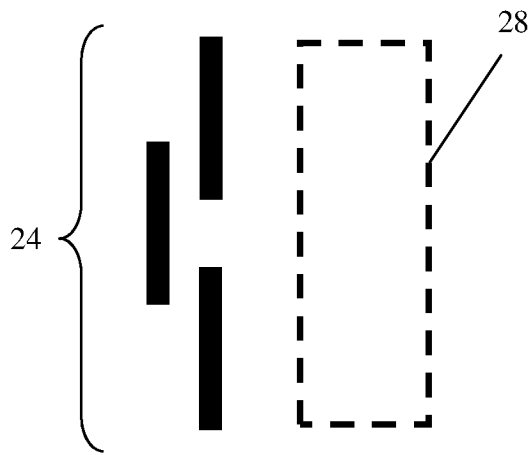
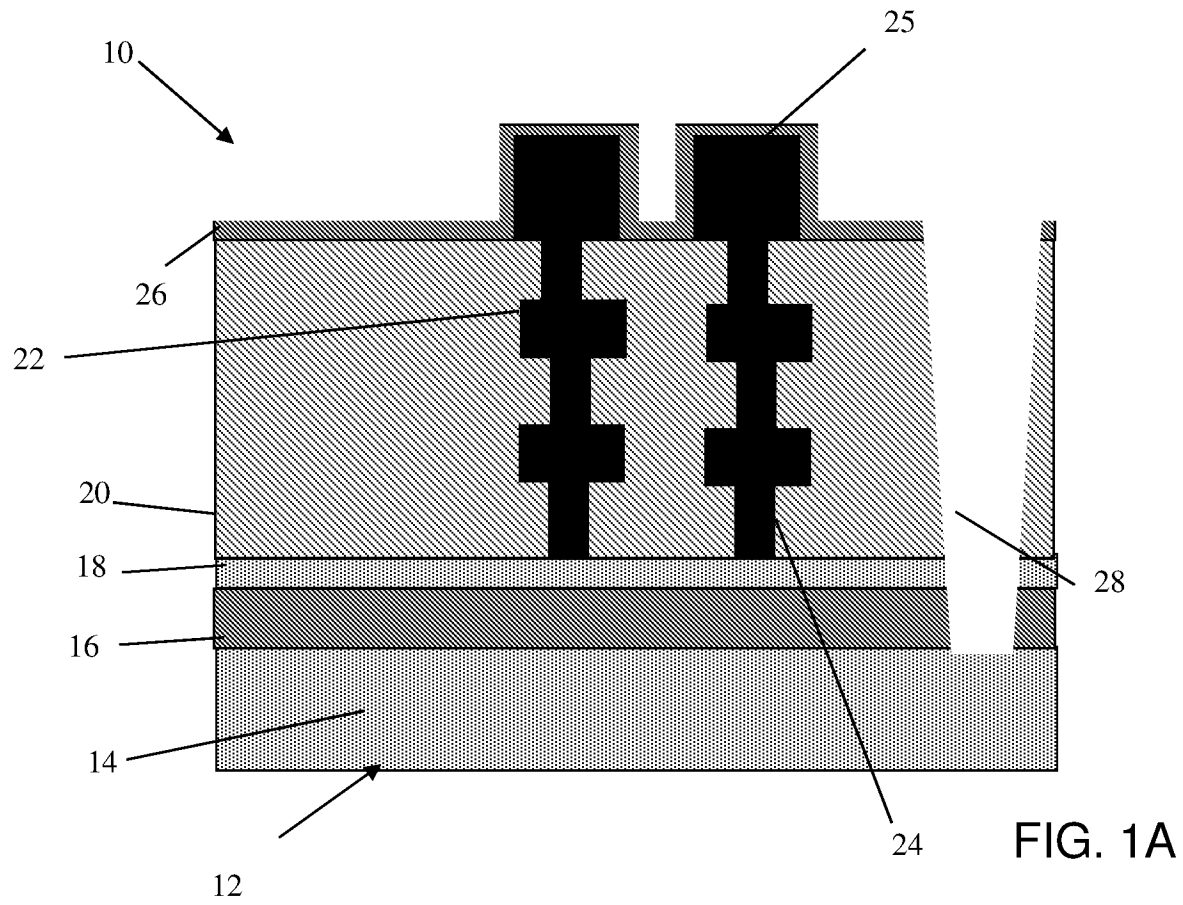
5C-5D). The airgap 50 can be partially or completely filled with oxide 110. The airgap 50 can be about 180 nm wide, formed by a pinch-off process as is known to those of skill in the art. For example, the airgap 50 can be formed by depositing nitride material 95 over the previously formed first level wires 105 that form both the crack stop and the other wires on the chip, followed by an oxide material 96 used to form wires and vias for the next wiring level. This nitride 95' and oxide 96 are deposited over the wires and crack stop formed in the low-k dielectric material 97. The oxide 96 and nitride 95 can be deposited using, e.g., PECVD processes. As shown in FIG. 5D, the airgap 50 be positioned partially over at least one side of a wiring structure 105', e.g., segmented guard ring structure. In this way, the air gaps 50 extending on both sides and between the wiring structure 105 forming an edge seal structure. In embodiments, the wiring structure 105 can be a contact for a first metal layer (e.g., M1 layer).

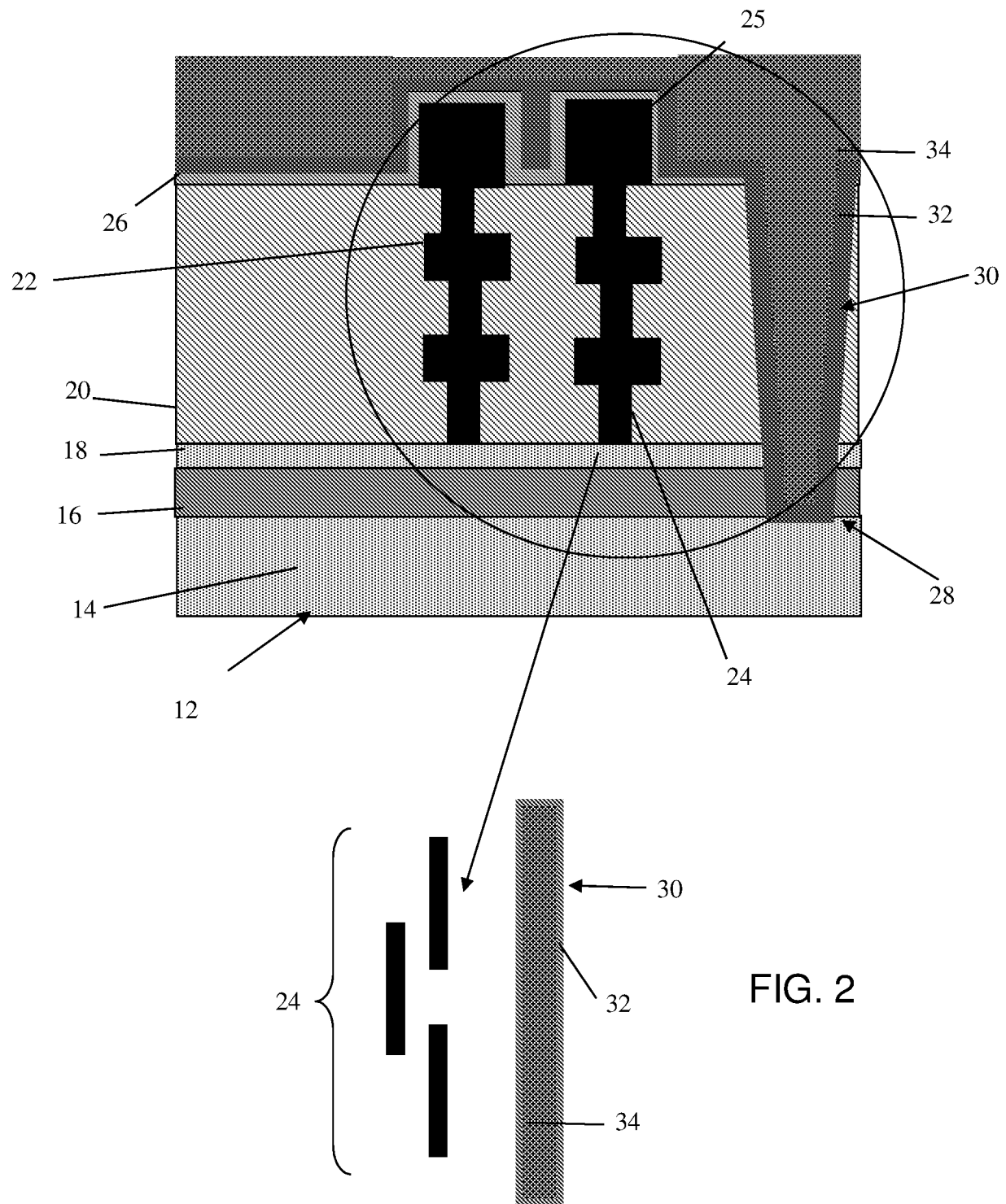
[0037] FIGS. 5A-5D show the low-k dielectric formed for the first wire level and contact below it. The low-k dielectric layer could be used for any one or more wire, contact, or via levels with the airgap extending completely through the low k dielectric layers. FIG. 5C shows the oxide 110 lined airgap 50 extending from the oxide layer 96 to the nitride layer 95' over substrate 98. FIG. 5D shows the oxide 110 lined airgap 50 landing partially on crack stop first metal layer 105' and also landing on nitride layer 95'. Again, the airgap is completely isolating the low k dielectric 97.

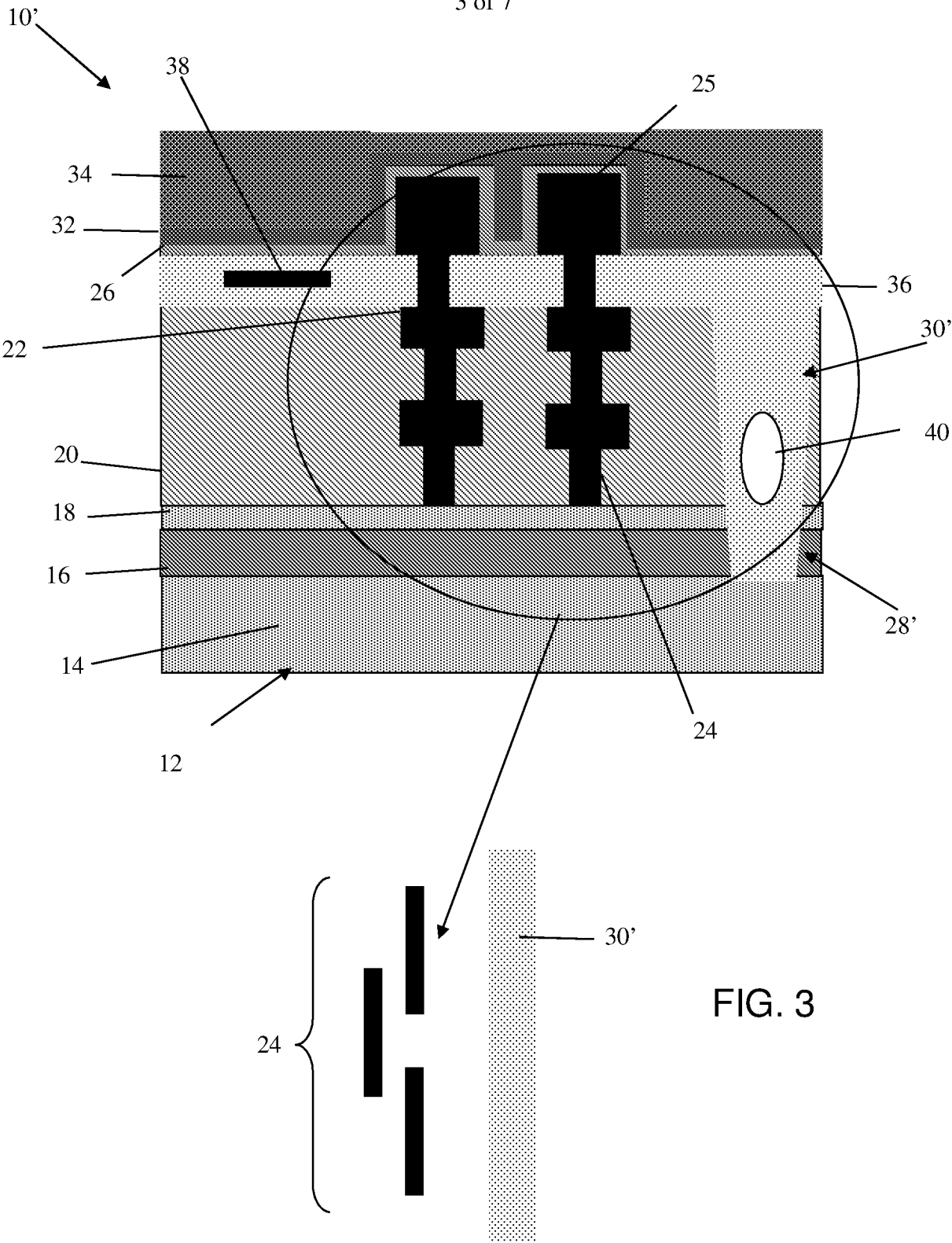
[0038] The method(s) as described above is used in the fabrication of integrated circuit chips. The resulting integrated circuit chips can be distributed by the fabricator in raw wafer form (that is, as a single wafer that has multiple unpackaged chips), as a bare die,

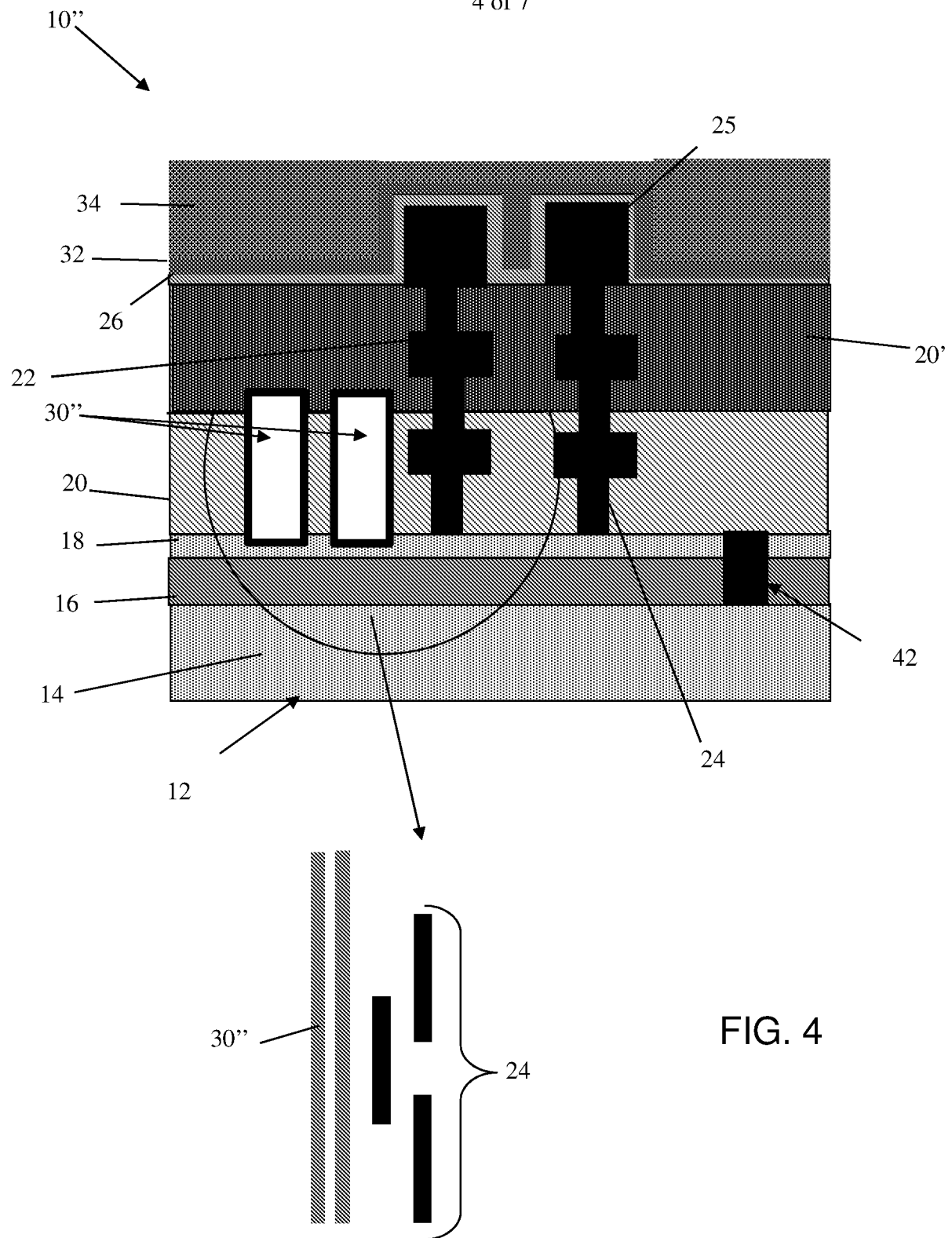
or in a packaged form. In the latter case the chip is mounted in a single chip package (such as a plastic carrier, with leads that are affixed to a motherboard or other higher level carrier) or in a multichip package (such as a ceramic carrier that has either or both surface interconnections or buried interconnections). In any case the chip is then integrated with other chips, discrete circuit elements, and/or other signal processing devices as part of either (a) an intermediate product, such as a motherboard, or (b) an end product. The end product can be any product that includes integrated circuit chips, ranging from toys and other low-end applications to advanced computer products having a display, a keyboard or other input device, and a central processor.

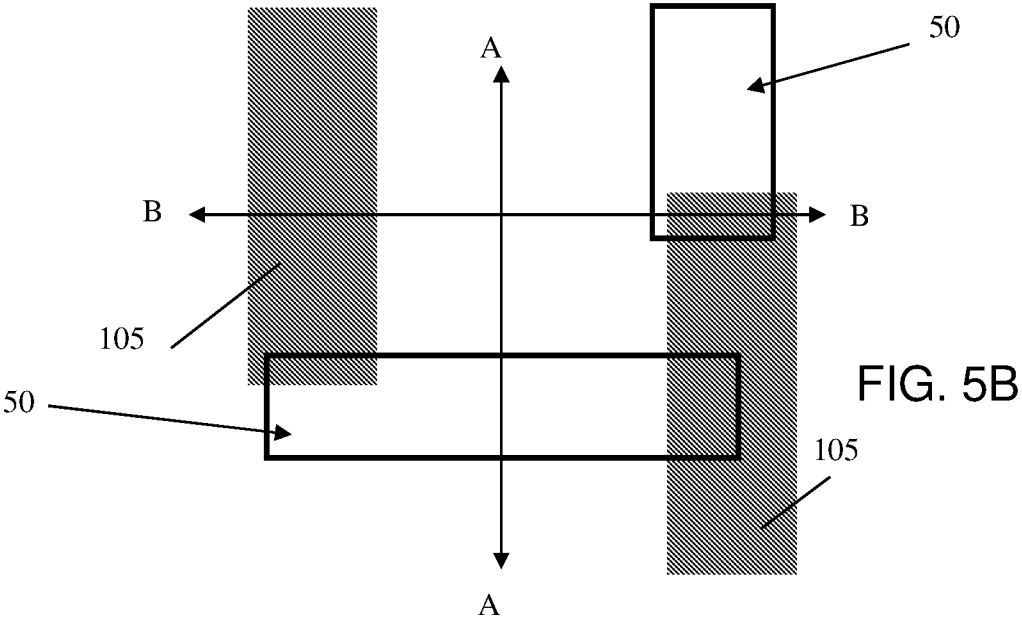
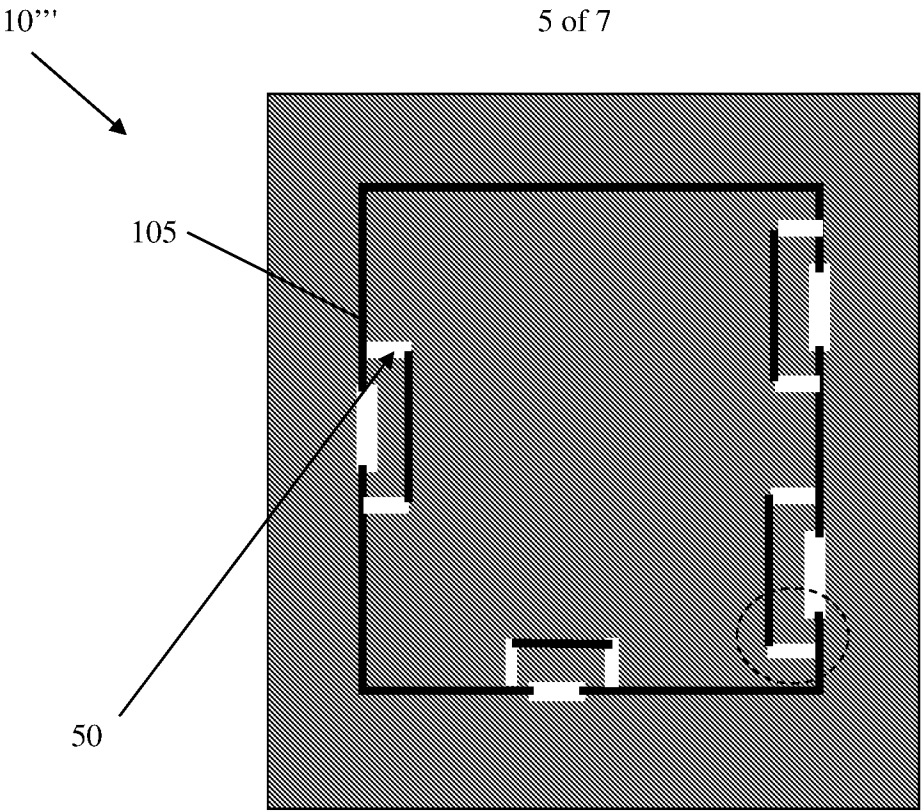
[0039] The descriptions of the various embodiments of the present disclosure have been presented for purposes of illustration, but are not intended to be exhaustive or limited to the embodiments disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the described embodiments. The terminology used herein was chosen to best explain the principles of the embodiments, the practical application or technical improvement over technologies found in the marketplace, or to enable others of ordinary skill in the art to understand the embodiments disclosed herein.











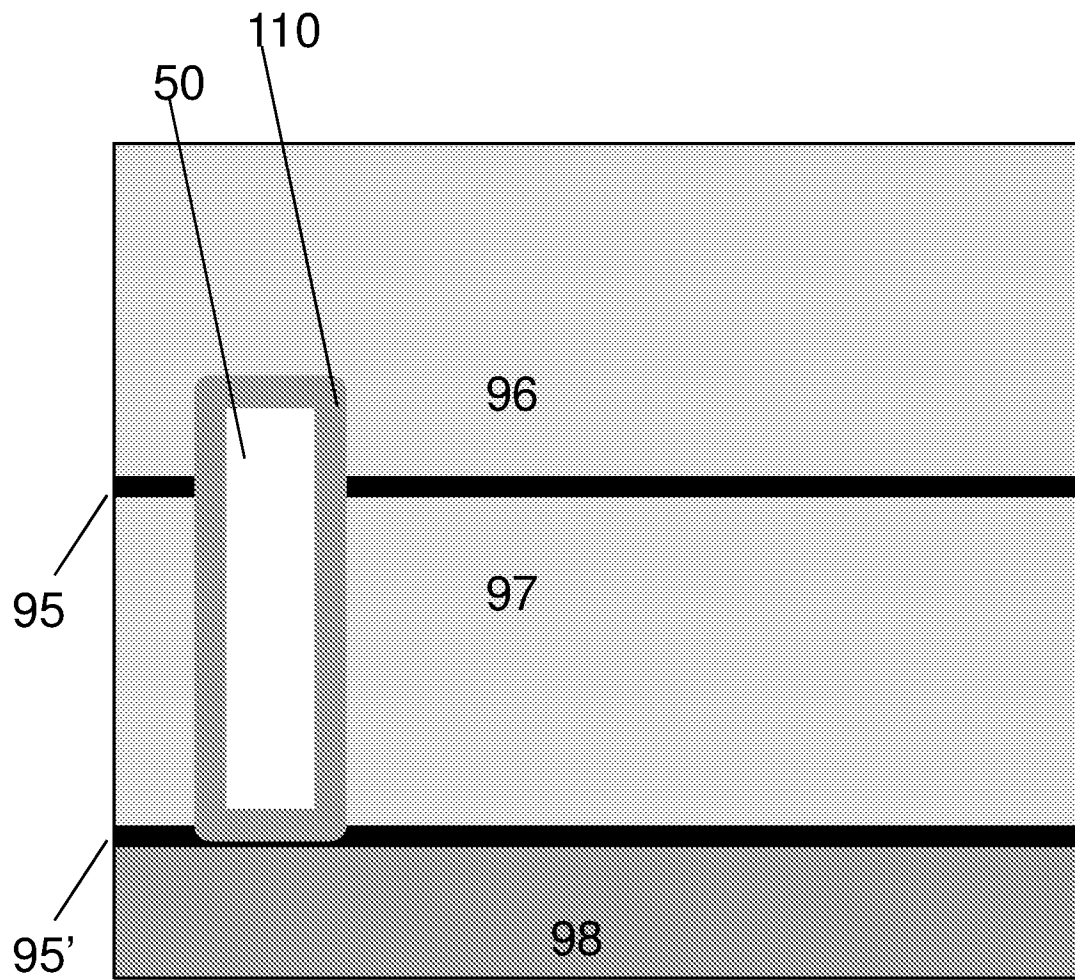


FIG. 5C

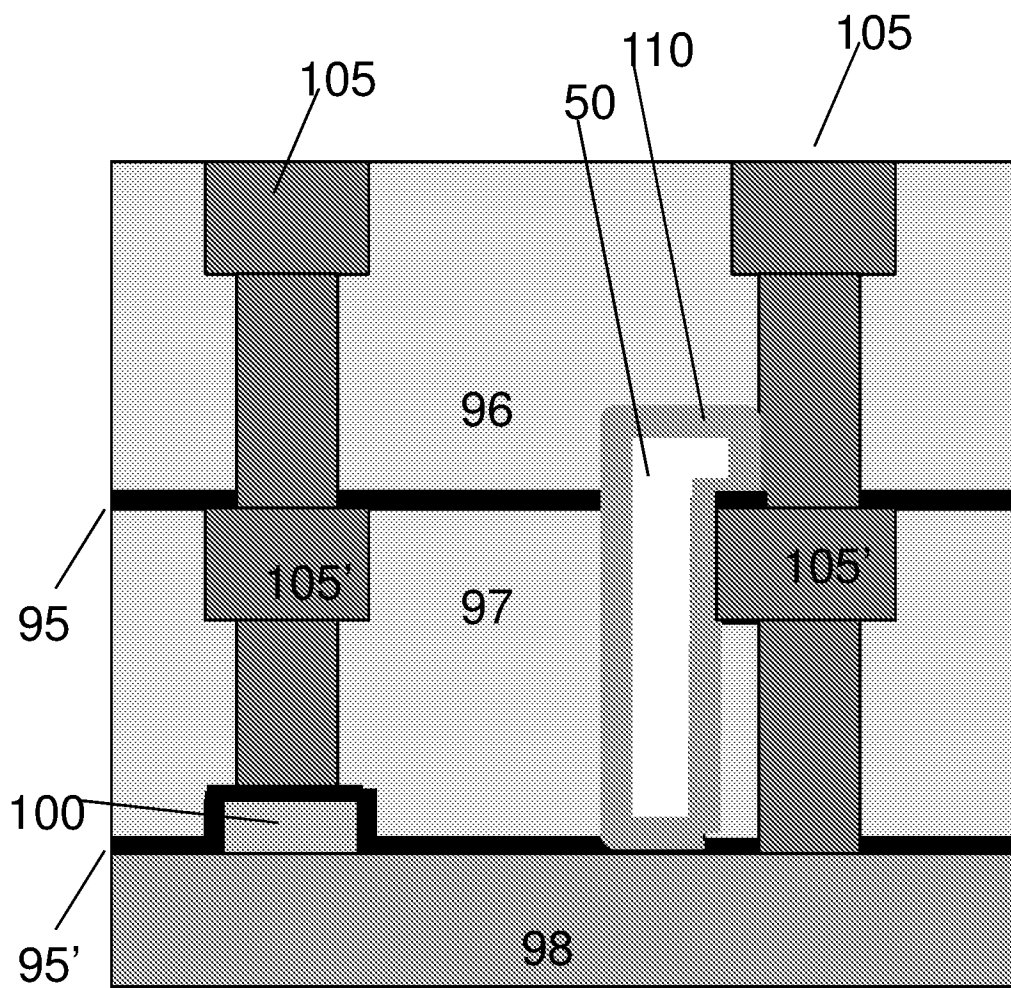


FIG. 5D

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	IBUR032-US-NP
		Application Number	
Title of Invention	SEGMENTED GUARD-RING AND CHIP EDGE SEALS		
The application data sheet is part of the provisional or nonprovisional application for which it is being submitted. The following form contains the bibliographic data arranged in a format specified by the United States Patent and Trademark Office as outlined in 37 CFR 1.76. This document may be completed electronically and submitted to the Office in electronic format using the Electronic Filing System (EFS) or the document may be printed and included in a paper filed application.			

Secrecy Order 37 CFR 5.2:

☐	Portions or all of the application associated with this Application Data Sheet may fall under a Secrecy Order pursuant to 37 CFR 5.2 (Paper filers only. Applications that fall under Secrecy Order may not be filed electronically.)
--------------------------	---

Inventor Information:

Inventor	1				Remove
Legal Name					
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	Anthony	K.	STAMPER		
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service					
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Inventor	2				Remove
Legal Name					
Prefix	Given Name	Middle Name	Family Name	Suffix	
	Vincent	J.	MCGAHAY		
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service					
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Postal Code	12533	Country	US		

Inventor	3				Remove
Legal Name					
Prefix	Given Name	Middle Name	Family Name	Suffix	
	Zhong-Xiang		HE		
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service					

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	IBUR032-US-NP
		Application Number	
Title of Invention	SEGMENTED GUARD-RING AND CHIP EDGE SEALS		

City	Essex Junction	State/Province	VT	Country of Residence	US
------	----------------	----------------	----	----------------------	----

Mailing Address of Inventor:

Address 1	1000 River Street				
Address 2					
City	Essex Junction	State/Province	VT		
Postal Code	05452	Country	US		
All Inventors Must Be Listed - Additional Inventor Information blocks may be generated within this form by selecting the Add button.					

Correspondence Information:

Enter either Customer Number or complete the Correspondence Information section below. For further information see 37 CFR 1.33(a).			
☐ An Address is being provided for the correspondence information of this application.			
Customer Number	133809		
Email Address		Add Email	Remove Email

Application Information:

Title of the Invention	SEGMENTED GUARD-RING AND CHIP EDGE SEALS		
Attorney Docket Number	IBUR032-US-NP	Small Entity Status Claimed	☐
Application Type	Nonprovisional		
Subject Matter	Utility		
Total Number of Drawing Sheets (if any)	7	Suggested Figure for Publication (if any)	1C

Filing By Reference:

Only complete this section when filing an application by reference under 35 U.S.C. 111(c) and 37 CFR 1.57(a). Do not complete this section if application papers including a specification and any drawings are being filed. Any domestic benefit or foreign priority information must be provided in the appropriate section(s) below (i.e., "Domestic Benefit/National Stage Information" and "Foreign Priority Information").

For the purposes of a filing date under 37 CFR 1.53(b), the description and any drawings of the present application are replaced by this reference to the previously filed application, subject to conditions and requirements of 37 CFR 1.57(a).

Application number of the previously filed application	Filing date (YYYY-MM-DD)	Intellectual Property Authority or Country

Publication Information:

☐ Request Early Publication (Fee required at time of Request 37 CFR 1.219)
☐ Request Not to Publish. I hereby request that the attached application not be published under 35 U.S.C. 122(b) and certify that the invention disclosed in the attached application has not and will not be the subject of an application filed in another country, or under a multilateral international agreement, that requires publication at eighteen months after filing.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	IBUR032-US-NP
		Application Number	
Title of Invention	SEGMENTED GUARD-RING AND CHIP EDGE SEALS		

Representative Information:

Representative information should be provided for all practitioners having a power of attorney in the application. Providing this information in the Application Data Sheet does not constitute a power of attorney in the application (see 37 CFR 1.32). Either enter Customer Number or complete the Representative Name section below. If both sections are completed the customer Number will be used for the Representative Information during processing.

Please Select One:	☒ Customer Number	☐ US Patent Practitioner	☐ Limited Recognition (37 CFR 11.9)
Customer Number	133809		

Domestic Benefit/National Stage Information:

This section allows for the applicant to either claim benefit under 35 U.S.C. 119(e), 120, 121, 365(c), or 386(c) or indicate National Stage entry from a PCT application. Providing benefit claim information in the Application Data Sheet constitutes the specific reference required by 35 U.S.C. 119(e) or 120, and 37 CFR 1.78.

When referring to the current application, please leave the "Application Number" field blank.

Prior Application Status		Remove
Application Number	Continuity Type	Prior Application Number
Additional Domestic Benefit/National Stage Data may be generated within this form by selecting the Add button.		Add

Foreign Priority Information:

This section allows for the applicant to claim priority to a foreign application. Providing this information in the application data sheet constitutes the claim for priority as required by 35 U.S.C. 119(b) and 37 CFR 1.55. When priority is claimed to a foreign application that is eligible for retrieval under the priority document exchange program (PDX)ⁱ the information will be used by the Office to automatically attempt retrieval pursuant to 37 CFR 1.55(i)(1) and (2). Under the PDX program, applicant bears the ultimate responsibility for ensuring that a copy of the foreign application is received by the Office from the participating foreign intellectual property office, or a certified copy of the foreign priority application is filed, within the time period specified in 37 CFR 1.55(g)(1).

Application Number	Country ⁱ	Filing Date (YYYY-MM-DD)	Access Code ⁱ (if applicable)
Additional Foreign Priority Data may be generated within this form by selecting the Add button.			Add

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	IBUR032-US-NP
		Application Number	
Title of Invention	SEGMENTED GUARD-RING AND CHIP EDGE SEALS		

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications

This application (1) claims priority to or the benefit of an application filed before March 16, 2013 and (2) also contains, or contained at any time, a claim to a claimed invention that has an effective filing date on or after March 16, 2013. ☐ NOTE: By providing this statement under 37 CFR 1.55 or 1.78, this application, with a filing date on or after March 16, 2013, will be examined under the first inventor to file provisions of the AIA.	
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Application Data Sheet 37 CFR 1.76		Attorney Docket Number	IBUR032-US-NP
		Application Number	
Title of Invention	SEGMENTED GUARD-RING AND CHIP EDGE SEALS		

Authorization or Opt-Out of Authorization to Permit Access:

When this Application Data Sheet is properly signed and filed with the application, applicant has provided written authority to permit a participating foreign intellectual property (IP) office access to the instant application-as-filed (see paragraph A in subsection 1 below) and the European Patent Office (EPO) access to any search results from the instant application (see paragraph B in subsection 1 below).

Should applicant choose not to provide an authorization identified in subsection 1 below, applicant **must opt-out** of the authorization by checking the corresponding box A or B or both in subsection 2 below.

NOTE: This section of the Application Data Sheet is **ONLY** reviewed and processed with the **INITIAL** filing of an application. After the initial filing of an application, an Application Data Sheet cannot be used to provide or rescind authorization for access by a foreign IP office(s). Instead, Form PTO/SB/39 or PTO/SB/69 must be used as appropriate.

1. Authorization to Permit Access by a Foreign Intellectual Property Office(s)

A. Priority Document Exchange (PDX) - Unless box A in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the European Patent Office (EPO), the Japan Patent Office (JPO), the Korean Intellectual Property Office (KIPO), the State Intellectual Property Office of the People's Republic of China (SIPO), the World Intellectual Property Organization (WIPO), and any other foreign intellectual property office participating with the USPTO in a bilateral or multilateral priority document exchange agreement in which a foreign application claiming priority to the instant patent application is filed, access to: (1) the instant patent application-as-filed and its related bibliographic data, (2) any foreign or domestic application to which priority or benefit is claimed by the instant application and its related bibliographic data, and (3) the date of filing of this Authorization. See 37 CFR 1.14(h)(1).

B. Search Results from U.S. Application to EPO - Unless box B in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the EPO access to the bibliographic data and search results from the instant patent application when a European patent application claiming priority to the instant patent application is filed. See 37 CFR 1.14(h)(2).

The applicant is reminded that the EPO's Rule 141(1) EPC (European Patent Convention) requires applicants to submit a copy of search results from the instant application without delay in a European patent application that claims priority to the instant application.

2. Opt-Out of Authorizations to Permit Access by a Foreign Intellectual Property Office(s)

☐ A. Applicant **DOES NOT** authorize the USPTO to permit a participating foreign IP office access to the instant application-as-filed. If this box is checked, the USPTO will not be providing a participating foreign IP office with any documents and information identified in subsection 1A above.

☐ B. Applicant **DOES NOT** authorize the USPTO to transmit to the EPO any search results from the instant patent application. If this box is checked, the USPTO will not be providing the EPO with search results from the instant application.

NOTE: Once the application has published or is otherwise publicly available, the USPTO may provide access to the application in accordance with 37 CFR 1.14.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	IBUR032-US-NP
		Application Number	
Title of Invention	SEGMENTED GUARD-RING AND CHIP EDGE SEALS		

Applicant Information:

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.

Applicant	1	Remove
If the applicant is the inventor (or the remaining joint inventor or inventors under 37 CFR 1.45), this section should not be completed. The information to be provided in this section is the name and address of the legal representative who is the applicant under 37 CFR 1.43; or the name and address of the assignee, person to whom the inventor is under an obligation to assign the invention, or person who otherwise shows sufficient proprietary interest in the matter who is the applicant under 37 CFR 1.46. If the applicant is an applicant under 37 CFR 1.46 (assignee, person to whom the inventor is obligated to assign, or person who otherwise shows sufficient proprietary interest) together with one or more joint inventors, then the joint inventor or inventors who are also the applicant should be identified in this section. Clear		
☒ Assignee	Legal Representative under 35 U.S.C. 117	Joint Inventor
Person to whom the inventor is obligated to assign.		Person who shows sufficient proprietary interest
If applicant is the legal representative, indicate the authority to file the patent application, the inventor is:		
Name of the Deceased or Legally Incapacitated Inventor:		
If the Applicant is an Organization check here. ☒		
Organization Name	GLOBALFOUNDRIES INC.	
Mailing Address Information For Applicant:		
Address 1	PO Box 309, Ugland House	
Address 2		
City	GRAND CAYMAN	State/Province
Country	KY	Postal Code
Phone Number		Fax Number
Email Address		
Additional Applicant Data may be generated within this form by selecting the Add button. Add		

Assignee Information including Non-Applicant Assignee Information:

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	IBUR032-US-NP
		Application Number	
Title of Invention	SEGMENTED GUARD-RING AND CHIP EDGE SEALS		

Assignee	1			
Complete this section if assignee information, including non-applicant assignee information, is desired to be included on the patent application publication. An assignee-applicant identified in the "Applicant Information" section will appear on the patent application publication as an applicant. For an assignee-applicant, complete this section only if identification as an assignee is also desired on the patent application publication.				
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If the Assignee or Non-Applicant Assignee is an Organization check here.				☐
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Address 2				
City		State/Province		
Country ⁱ		Postal Code		
Phone Number		Fax Number		
Email Address				
Additional Assignee or Non-Applicant Assignee Data may be generated within this form by selecting the Add button.				Add

Signature:

NOTE: This Application Data Sheet must be signed in accordance with 37 CFR 1.33(b). However, if this Application Data Sheet is submitted with the **INITIAL** filing of the application and either box A or B is not checked in subsection 2 of the "Authorization or Opt-Out of Authorization to Permit Access" section, then this form must also be signed in accordance with 37 CFR 1.14(c).

This Application Data Sheet **must** be signed by a patent practitioner if one or more of the applicants is a **juristic entity** (e.g., corporation or association). If the applicant is two or more joint inventors, this form must be signed by a patent practitioner, **all** joint inventors who are the applicant, or one or more joint inventor-applicants who have been given power of attorney (e.g., see USPTO Form PTO/AIA/81) on behalf of **all** joint inventor-applicants.

See 37 CFR 1.4(d) for the manner of making signatures and certifications.

Signature	/Andrew M. Calderon/		Date (YYYY-MM-DD)	2017-02-27
First Name	Andrew M.	Last Name	Calderon	Registration Number
				38093
Additional Signature may be generated within this form by selecting the Add button.				Add

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	IBUR032-US-NP
		Application Number	
Title of Invention	SEGMENTED GUARD-RING AND CHIP EDGE SEALS		

This collection of information is required by 37 CFR 1.76. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 23 minutes to complete, including gathering, preparing, and submitting the completed application data sheet form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Electronic Patent Application Fee Transmittal				
Application Number:				
Filing Date:				
Title of Invention:		SEGMENTED GUARD-RING AND CHIP EDGE SEALS		
First Named Inventor/Applicant Name:		Anthony K. STAMPER		
Filer:		Andrew Michael Calderon/Aleksandra Paredes		
Attorney Docket Number:		IBUR032-US-NP		
Filed as Large Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
UTILITY APPLICATION FILING	1011	1	280	280
UTILITY SEARCH FEE	1111	1	600	600
UTILITY EXAMINATION FEE	1311	1	720	720
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				1600

Electronic Acknowledgement Receipt

EFS ID:	28468149
Application Number:	15443276
International Application Number:	
Confirmation Number:	2522
Title of Invention:	SEGMENTED GUARD-RING AND CHIP EDGE SEALS
First Named Inventor/Applicant Name:	Anthony K. STAMPER
Customer Number:	133809
Filer:	Andrew Michael Calderon/Aleksandra Paredes
Filer Authorized By:	Andrew Michael Calderon
Attorney Docket Number:	IBUR032-US-NP
Receipt Date:	27-FEB-2017
Filing Date:	
Time Stamp:	14:59:52
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	CARD
Payment was successfully received in RAM	\$ 1600
RAM confirmation Number	022817INTEFSW15004801
Deposit Account	502478
Authorized User	Andrew Calderon

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

37 CFR 1.16 (National application filing, search, and examination fees)

37 CFR 1.17 (Patent application and reexamination processing fees)

File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Miscellaneous Incoming Letter	IBUR032-US-NP_AuthorizationtoCharge.pdf	51792	no	1
			a53a472f6d8b1b60b15915403931baea44e2acb2		
Warnings:					
Information:					
2	Oath or Declaration filed	IBUR032-US-NP_FiledAssignDec_02-27-17.pdf	192682	no	3
			bc2287058208ea2a2f35d846385ea075e626891a		
Warnings:					
Information:					
3	Power of Attorney	IBUR032-US-NP_TransmittalandPOA.pdf	167890	no	2
			65348f824babbb6f1be2085b2f7d8bc1fe34440c		
Warnings:					
Information:					
4		IBUR032-US-NP_Application.pdf	75226	yes	20
			46a4aff3b6e340ec172e194132fb4c64eabe6f56		
	Multipart Description/PDF files in .zip description				
	Document Description		Start	End	
	Abstract		20	20	
	Claims		16	19	
	Specification		1	15	
Warnings:					
Information:					
5	Drawings-only black and white line drawings	IBUR032-US-NP_FormalDrawings.pdf	54508	no	7
			4e869abb1e895f99d04dea4035675bb43d4b2158		

Warnings:					
Information:					
6	Application Data Sheet	IBUR032-US-NP_ADS.pdf	1793139	no	9
			750db7c7042ac2623535ea2dab16e11e54bd4a9d		
Warnings:					
Information:					
7	Fee Worksheet (SB06)	fee-info.pdf	34898	no	2
			32c6d44a8e31f1a219524befabd017ab55d4102f		
Warnings:					
Information:					
Total Files Size (in bytes):			2370135		
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Inventor(s) : Anthony K. STAMPER et al. Group Art Unit : Not Yet Assigned
Appln. No. : Not Yet Assigned Examiner : Not Yet Assigned
Filed : Concurrently herewith Confirmation No. : Not Yet Assigned
For : SEGMENTED GUARD-RING AND CHIP EDGE SEALS

AUTHORIZATION TO CHARGE DEPOSIT ACCOUNT

Commissioner for Patents
U.S. Patent and Trademark Office
Customer Service Window, Mail Stop _____
Randolph Building
401 Dulany Street
Alexandria, VA 22314

Sir:

In accordance with 37 C.F.R. § 1.25, the undersigned hereby authorizes the U.S. Patent and Trademark Office to charge all fees set forth in 37 C.F.R. §§ 1.16 to 1.18 and §41.20 during the entire pendency of the above-identified application, including any basic filing fees, application size fees, search fees, examination fees, extension of time fees, excess claim fees, information disclosure statement fees, and any and all fees related to appeals including notice of appeal fees, appeal forwarding fees, and oral hearing fees, to Deposit Account No. 50-2478.

Respectfully submitted,



Andrew M. Calderon
Reg. No. 38,093

Roberts, Mlotkowski, Safran, Cole & Calderon PC
P.O. Box 10064
McLean, VA 22102
Phone: 703.677.3011
Fax: 703.848.2981
Email: acalderon@rmssc2.com

**COMBINED DECLARATION and ASSIGNMENT (37 CFR 1.63) FOR UTILITY OR DESIGN APPLICATION
USING AN APPLICATION DATA SHEET (37 CFR 1.76)****Title of
Invention**

SEGMENTED GUARD-RING AND CHIP EDGE SEALS

DECLARATION

As the below named inventor, I hereby declare that:

This declaration
is directed to:

The attached application, or

United States application or PCT international application number _____ filed on

The above-identified application was made or authorized to be made by me.

I believe that I am the original inventor or an original joint inventor of a claimed invention in the application.

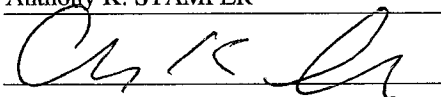
I hereby acknowledge that any willful false statement made in this declaration is punishable under 18 U.S.C. 1001 by fine or imprisonment of not more than five (5) years, or both.

I have reviewed and understand the contents of the application and am aware of the duty to disclose to the US Patent and Trademark Office all information known to me to be material to patentability as defined in 37 CFR § 1.56.

ASSIGNMENT

For good and valuable consideration, the receipt, sufficiency and adequacy of which are hereby acknowledged, I hereby sell, assign, transfer, to GLOBALFOUNDRIES Inc. having a business address of PO BOX 309, UGLAND HOUSE, GRAND CAYMAN, CAYMAN ISLANDS KY1-1104 its successors and assigns, absolutely and forever, my entire right, title and interest in and to the invention listed above, together with the Application, any and all Patents that may issue in the United States and/or any foreign countries, and any reissues, re-examinations, renewals, continuations, continuation-in-parts, divisionals or extensions thereof that may be issued or granted, and all right, title and interest to the inventions contained in said Patents and Applications, for the United States and all foreign countries, and all the rights and privileges relating thereto including but not limited to the priority rights arising from said Applications, the right to recover and take all such proceedings as may be necessary for the recovery of damages or otherwise in respect of past, present and future infringement, and the right to apply for, take and maintain patents on said inventions.

If any portion of this Combined Declaration or Assignment is held unenforceable, then remaining portions hereof shall nevertheless remain in full force and effect.

LEGAL NAME OF INVENTORInventor: Anthony K. STAMPERDate: Feb. 23, 2017Signature: 

Note: An application data sheet (PTO/SB/14 or equivalent), including naming the entire inventive entity, must accompany this form. Use an additional form for each additional inventor.

**COMBINED DECLARATION and ASSIGNMENT (37 CFR 1.63) FOR UTILITY OR DESIGN APPLICATION
USING AN APPLICATION DATA SHEET (37 CFR 1.76)**

**Title of
Invention**

SEGMENTED GUARD-RING AND CHIP EDGE SEALS

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is directed to:



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ASSIGNMENT

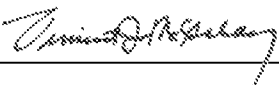
For good and valuable consideration, the receipt, sufficiency and adequacy of which are hereby acknowledged, I hereby sell, assign, transfer, to GLOBALFOUNDRIES Inc. having a business address of PO BOX 309, UGLAND HOUSE, GRAND CAYMAN, CAYMAN ISLANDS KY1-1104 its successors and assigns, absolutely and forever, my entire right, title and interest in and to the invention listed above, together with the Application, any and all Patents that may issue in the United States and/or any foreign countries, and any reissues, re-examinations, renewals, continuations, continuation-in-parts, divisionals or extensions thereof that may be issued or granted, and all right, title and interest to the inventions contained in said Patents and Applications, for the United States and all foreign countries, and all the rights and privileges relating thereto including but not limited to the priority rights arising from said Applications, the right to recover and take all such proceedings as may be necessary for the recovery of damages or otherwise in respect of past, present and future infringement, and the right to apply for, take and maintain patents on said inventions.

If any portion of this Combined Declaration or Assignment is held unenforceable, then remaining portions hereof shall nevertheless remain in full force and effect.

LEGAL NAME OF INVENTOR

Inventor: Vincent J. MCGAHAY

Date : February 24, 2017

Signature: 

Note: An application data sheet (PTO/SB/14 or equivalent), including naming the entire inventive entity, must accompany this form. Use an additional form for each additional inventor.

**COMBINED DECLARATION and ASSIGNMENT (37 CFR 1.63) FOR UTILITY OR DESIGN APPLICATION
USING AN APPLICATION DATA SHEET (37 CFR 1.76)****Title of
Invention**

SEGMENTED GUARD-RING AND CHIP EDGE SEALS

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This declaration
is directed to:

The attached application, or

United States application or PCT international application number _____ filed on

The above-identified application was made or authorized to be made by me.

I believe that I am the original inventor or an original joint inventor of a claimed invention in the application.

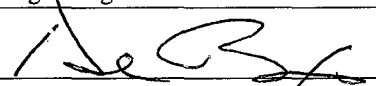
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If any portion of this Combined Declaration or Assignment is held unenforceable, then remaining portions hereof shall nevertheless remain in full force and effect.

LEGAL NAME OF INVENTORInventor: Zhong-Xiang HEDate : 02/23/2017Signature: 

Note: An application data sheet (PTO/SB/14 or equivalent), including naming the entire inventive entity, must accompany this form. Use an additional form for each additional inventor.

SCORE Placeholder Sheet for IFW Content

Application Number: 15443276

Document Date: 02/27/2017

The presence of this form in the IFW record indicates that the following document type was received in electronic format on the date identified above. This content is stored in the SCORE database.

- Drawings – Other than Black and White Line Drawings

Since this was an electronic submission, there is no physical artifact folder, no artifact folder is recorded in PALM, and no paper documents or physical media exist. The TIFF images in the IFW record were created from the original documents that are stored in SCORE.

To access the documents in the SCORE database, refer to instructions below.

At the time of document entry (noted above):

- Examiners may access SCORE content via the eDAN interface.
- Other USPTO employees can bookmark the current SCORE URL (<http://Score.uspto.gov/ScoreAccessWeb/>).
- External customers may access SCORE content via the Public and Private PAIR interfaces.

PATENT APPLICATION FEE DETERMINATION RECORD

Substitute for Form PTO-875

Application or Docket Number
15/443,276

APPLICATION AS FILED - PART I

(Column 1)		(Column 2)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
FOR	NUMBER FILED	NUMBER EXTRA	RATE(\$)	FEE(\$)		RATE(\$)	FEE(\$)
BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A			N/A	280
SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A	N/A			N/A	600
EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A			N/A	720
TOTAL CLAIMS (37 CFR 1.16(j))	20 minus 20 =	*			OR	x 80 =	0.00
INDEPENDENT CLAIMS (37 CFR 1.16(h))	3 minus 3 =	*				x 420 =	0.00
APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).						0.00
MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))							0.00
* If the difference in column 1 is less than zero, enter "0" in column 2.			TOTAL			TOTAL	1600

APPLICATION AS AMENDED - PART II

(Column 1)		(Column 2)	(Column 3)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
AMENDMENT A	CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE(\$)	ADDITIONAL FEE(\$)		RATE(\$)	ADDITIONAL FEE(\$)
Total (37 CFR 1.16(i))	*	Minus	**	=		OR	x	=
Independent (37 CFR 1.16(h))	*	Minus	***	=		OR	x	=
Application Size Fee (37 CFR 1.16(s))						OR		
FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						OR		
				TOTAL ADD'L FEE		OR	TOTAL ADD'L FEE	

(Column 1)		(Column 2)	(Column 3)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
AMENDMENT B	CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE(\$)	ADDITIONAL FEE(\$)		RATE(\$)	ADDITIONAL FEE(\$)
Total (37 CFR 1.16(i))	*	Minus	**	=		OR	x	=
Independent (37 CFR 1.16(h))	*	Minus	***	=		OR	x	=
Application Size Fee (37 CFR 1.16(s))						OR		
FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						OR		
				TOTAL ADD'L FEE		OR	TOTAL ADD'L FEE	

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".

*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".

The "Highest Number Previously Paid For" (Total or Independent) is the highest found in the appropriate box in column 1.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
15/443,276	02/27/2017	2811	1600	IBUR032-US-NP	20	3

CONFIRMATION NO. 2522

FILING RECEIPT



0000000089722922

133809
Roberts Mlotkowski Safran Cole & Calderon, P.C.
Intellectual Property Department
P.O. Box 10064
McLean, VA 22102-8064

Date Mailed: 03/09/2017

Receipt is acknowledged of this non-provisional patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF APPLICANT, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection. Please verify the accuracy of the data presented on this receipt. **If an error is noted on this Filing Receipt, please submit a written request for a Filing Receipt Correction. Please provide a copy of this Filing Receipt with the changes noted thereon. If you received a "Notice to File Missing Parts" for this application, please submit any corrections to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections**

Inventor(s)

Anthony K. STAMPER, Burlington, VT;
Vincent J. MCGAHAY, Poughkeepsie, NY;
Zhong-Xiang HE, Essex Junction, VT;

Applicant(s)

GLOBALFOUNDRIES INC., GRAND CAYMAN, CAYMAN ISLANDS;

Power of Attorney: The patent practitioners associated with Customer Number 133809

Domestic Applications for which benefit is claimed - None.

A proper domestic benefit claim must be provided in an Application Data Sheet in order to constitute a claim for domestic benefit. See 37 CFR 1.76 and 1.78.

Foreign Applications for which priority is claimed (You may be eligible to benefit from the **Patent Prosecution Highway** program at the USPTO. Please see <http://www.uspto.gov> for more information.) - None.

Foreign application information must be provided in an Application Data Sheet in order to constitute a claim to foreign priority. See 37 CFR 1.55 and 1.76.

Permission to Access Application via Priority Document Exchange: Yes

Permission to Access Search Results: Yes

Applicant may provide or rescind an authorization for access using Form PTO/SB/39 or Form PTO/SB/69 as appropriate.

If Required, Foreign Filing License Granted: 03/07/2017

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 15/443,276**

Projected Publication Date: 08/30/2018

Non-Publication Request: No

Early Publication Request: No

Title

SEGMENTED GUARD-RING AND CHIP EDGE SEALS

Preliminary Class

257

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications: No

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

Applicants also are advised that in the case of inventions made in the United States, the Director of the USPTO must issue a license before applicants can apply for a patent in a foreign country. The filing of a U.S. patent application serves as a request for a foreign filing license. The application's filing receipt contains further information and guidance as to the status of applicant's license for foreign filing.

Applicants may wish to consult the USPTO booklet, "General Information Concerning Patents" (specifically, the section entitled "Treaties and Foreign Patents") for more information on timeframes and deadlines for filing foreign patent applications. The guide is available either by contacting the USPTO Contact Center at 800-786-9199, or it can be viewed on the USPTO website at <http://www.uspto.gov/web/offices/pac/doc/general/index.html>.

For information on preventing theft of your intellectual property (patents, trademarks and copyrights), you may wish to consult the U.S. Government website, <http://www.stopfakes.gov>. Part of a Department of Commerce initiative, this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4258).

LICENSE FOR FOREIGN FILING UNDER
Title 35, United States Code, Section 184
Title 37, Code of Federal Regulations, 5.11 & 5.15

GRANTED

The applicant has been granted a license under 35 U.S.C. 184, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" followed by a date appears on this form. Such licenses are issued in all applications where the conditions for issuance of a license have been met, regardless of whether or not a license may be required as set forth in 37 CFR 5.15. The scope and limitations of this license are set forth in 37 CFR 5.15(a) unless an earlier license has been issued under 37 CFR 5.15(b). The license is subject to revocation upon written notification. The date indicated is the effective date of the license, unless an earlier license of similar scope has been granted under 37 CFR 5.13 or 5.14.

This license is to be retained by the licensee and may be used at any time on or after the effective date thereof unless it is revoked. This license is automatically transferred to any related applications(s) filed under 37 CFR 1.53(d). This license is not retroactive.

The grant of a license does not in any way lessen the responsibility of a licensee for the security of the subject matter as imposed by any Government contract or the provisions of existing laws relating to espionage and the national security or the export of technical data. Licensees should apprise themselves of current regulations especially with respect to certain countries, of other agencies, particularly the Office of Defense Trade Controls, Department of State (with respect to Arms, Munitions and Implements of War (22 CFR 121-128)); the Bureau of Industry and Security, Department of Commerce (15 CFR parts 730-774); the Office of Foreign Assets Control, Department of Treasury (31 CFR Parts 500+) and the Department of Energy.

NOT GRANTED

No license under 35 U.S.C. 184 has been granted at this time, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" DOES NOT appear on this form. Applicant may still petition for a license under 37 CFR 5.12, if a license is desired before the expiration of 6 months from the filing date of the application. If 6 months has lapsed from the filing date of this application and the licensee has not received any indication of a secrecy order under 35 U.S.C. 181, the licensee may foreign file the application pursuant to 37 CFR 5.15(b).

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The United States represents the largest, most dynamic marketplace in the world and is an unparalleled location for business investment, innovation, and commercialization of new technologies. The U.S. offers tremendous resources and advantages for those who invest and manufacture goods here. Through SelectUSA, our nation works to promote and facilitate business investment. SelectUSA provides information assistance to the international investor community; serves as an ombudsman for existing and potential investors; advocates on behalf of U.S. cities, states, and regions competing for global investment; and counsels U.S. economic development organizations on investment attraction best practices. To learn more about why the United States is the best country in the world to develop technology, manufacture products, deliver services, and grow your business, visit <http://www.SelectUSA.gov> or call +1-202-482-6800.



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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
15/443,276	02/27/2017	Anthony K. STAMPER	IBUR032-US-NP	2522
133809	7590	07/24/2017	EXAMINER	
Roberts Mlotkowski Safran Cole & Calderon, P.C.			PHAM, LONG	
Intellectual Property Department			ART UNIT	
P.O. Box 10064			PAPER NUMBER	
McLean, VA 22102-8064			2814	
			NOTIFICATION DATE	
			DELIVERY MODE	
			07/24/2017	
			ELECTRONIC	

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

docketing@rmc2.com
lgalllaugh@rmc2.com
dbeltran@rmc2.com

Office Action Summary	Application No. 15/443,276	Applicant(s) STAMPER ET AL.	
	Examiner LONG PHAM	Art Unit 2814	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 2 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

- 1) ☐ Responsive to communication(s) filed on _____.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.
- 2a) ☐ This action is **FINAL**. 2b) ☐ This action is non-final.
- 3) ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.
- 4) ☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

- 5) ☒ Claim(s) 1-20 is/are pending in the application.
5a) Of the above claim(s) _____ is/are withdrawn from consideration.
- 6) ☐ Claim(s) _____ is/are allowed.
- 7) ☐ Claim(s) _____ is/are rejected.
- 8) ☐ Claim(s) _____ is/are objected to.
- 9) ☒ Claim(s) 1-20 are subject to restriction and/or election requirement.

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

Application Papers

- 10) ☐ The specification is objected to by the Examiner.
- 11) ☐ The drawing(s) filed on _____ is/are: a) ☐ accepted or b) ☐ objected to by the Examiner.
Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).
Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

- 12) ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

- a) ☐ All b) ☐ Some** c) ☐ None of the:
1. ☐ Certified copies of the priority documents have been received.
 2. ☐ Certified copies of the priority documents have been received in Application No. _____.
 3. ☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

- | | |
|---|---|
| 1) ☐ Notice of References Cited (PTO-892) | 3) ☐ Interview Summary (PTO-413)
Paper No(s)/Mail Date. _____ |
| 2) ☐ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b)
Paper No(s)/Mail Date _____ | 4) ☐ Other: _____ |

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

DETAILED ACTION

Election/Restrictions

Restriction to one of the following inventions is required under 35 U.S.C. 121:

- I. Claim 20, drawn to a process, classified in H01L2924/00.
- II. Claims 1-10 and 12-19, drawn to a device, classified in H01L2224/73265.

The inventions are distinct, each from the other because of the following reasons:

Inventions I and II are related as process of making and product made. The inventions are distinct if either or both of the following can be shown: (1) that the process as claimed can be used to make another and materially different product or (2) that the product as claimed can be made by another and materially different process (MPEP § 806.05(f)). In the instant case, the product as claimed can be made by another and materially different process such as one in which the wiring structure and guard ring can be formed at the same time.

Restriction for examination purposes as indicated is proper because all the inventions listed in this action are independent or distinct for the reasons given above and there would be a serious search and/or examination burden if restriction were not required because one or more of the following reasons apply:

The searches for the device and process inventions are not co-extensive.

Applicant is advised that the reply to this requirement to be complete must include (i) an election of a invention to be examined even though the requirement may be traversed (37 CFR 1.143) and (ii) identification of the claims encompassing the elected invention.

The election of an invention may be made with or without traverse. To reserve a right to petition, the election must be made with traverse. If the reply does not distinctly and specifically point out supposed errors in the restriction requirement, the election shall be treated as an election without traverse. Traversal must be presented at the time of election in order to be considered timely. Failure to timely traverse the requirement will result in the loss of right to petition under 37 CFR 1.144. If claims are added after the election, applicant must indicate which of these claims are readable upon the elected invention.

Should applicant traverse on the ground that the inventions are not patentably distinct, applicant should submit evidence or identify such evidence now of record showing the inventions to be obvious variants or clearly admit on the record that this is the case. In either instance, if the examiner finds one of the inventions unpatentable over the prior art, the evidence or admission may be used in a rejection under 35 U.S.C. 103 or pre-AIA 35 U.S.C. 103(a) of the other invention.

This application contains claims directed to the following patentably distinct species: species or embodiment 1 para 0015, species or embodiment 2 of para 0016. The species are independent or distinct because **the patentability or unpatentability of one species does not necessarily imply the patentability or unpatentability of another species**. In addition, these species are not obvious variants of each other based on the current record.

Applicant is required under 35 U.S.C. 121 to elect a single disclosed species, or a single grouping of patentably indistinct species, for prosecution on the merits to which the claims shall be restricted if no generic claim is finally held to be allowable. Currently, **claim 1 appears to be generic to claim 12**.

There is a search and/or examination burden for the patentably distinct species as set forth above because at least the following reason(s)

apply: **The searches of the patentably distinct species are not co-extensive.**

Applicant is advised that the reply to this requirement to be complete must include (i) an election of a species to be examined even though the requirement may be traversed (37 CFR 1.143) **and (ii) identification of the claims encompassing the elected species or grouping of patentably indistinct species**, including any claims subsequently added. An argument that a claim is allowable or that all claims are generic is considered nonresponsive unless accompanied by an election.

The election may be made with or without traverse. To preserve a right to petition, the election must be made with traverse. If the reply does not distinctly and specifically point out supposed errors in the election of species requirement, the election shall be treated as an election without traverse. Traversal must be presented at the time of election in order to be considered timely. Failure to timely traverse the requirement will result in the loss of right to petition under 37 CFR 1.144. If claims are added after the election, applicant must indicate which of these claims are readable on the elected species or grouping of patentably indistinct species.

Should applicant traverse on the ground that the species, or groupings of patentably indistinct species from which election is required, are not patentably distinct, applicant should submit evidence or identify such

evidence now of record showing them to be obvious variants or clearly admit on the record that this is the case. In either instance, if the examiner finds one of the species unpatentable over the prior art, the evidence or admission may be used in a rejection under 35 U.S.C. 103 or pre-AIA 35 U.S.C. 103(a) of the other species.

Upon the allowance of a generic claim, applicant will be entitled to consideration of claims to additional species which depend from or otherwise require all the limitations of an allowable generic claim as provided by 37 CFR 1.141. A telephone call was made to -- on -- to request an oral election to the above restriction requirement, but did not result in an election being made.

Applicant is reminded that upon the cancellation of claims to a non-elected invention, the inventorship must be corrected in compliance with 37 CFR 1.48(a) if one or more of the currently named inventors is no longer an inventor of at least one claim remaining in the application. A request to correct inventorship under 37 CFR 1.48(a) must be accompanied by an application data sheet in accordance with 37 CFR 1.76 that identifies each inventor by his or her legal name and by the processing fee required under 37 CFR 1.17(i).

The examiner has required restriction between product or apparatus claims and process claims. Where applicant elects claims directed to the

product/apparatus, and all product/apparatus claims are subsequently found allowable, withdrawn process claims that include all the limitations of the allowable product/apparatus claims should be considered for rejoinder. All claims directed to a nonelected process invention must include all the limitations of an allowable product/apparatus claim for that process invention to be rejoined.

In the event of rejoinder, the requirement for restriction between the product/apparatus claims and the rejoined process claims will be withdrawn, and the rejoined process claims will be fully examined for patentability in accordance with 37 CFR 1.104. Thus, to be allowable, the rejoined claims must meet all criteria for patentability including the requirements of 35 U.S.C. 101, 102, 103 and 112. Until all claims to the elected product/apparatus are found allowable, an otherwise proper restriction requirement between product/apparatus claims and process claims may be maintained. Withdrawn process claims that are not commensurate in scope with an allowable product/apparatus claim will not be rejoined. See MPEP § 821.04. Additionally, in order for rejoinder to occur, applicant is advised that the process claims should be amended during prosecution to require the limitations of the product/apparatus claims. **Failure to do so may result in no rejoinder.** Further, note that the prohibition against double patenting

Art Unit: 2814

rejections of 35 U.S.C. 121 does not apply where the restriction requirement is withdrawn by the examiner before the patent issues. See MPEP § 804.01.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to LONG PHAM whose telephone number is (571)272-1714. The examiner can normally be reached on Mon-Frid, 10am to 5pm.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at <http://www.uspto.gov/interviewpractice>.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Wael Fahmy can be reached on 571-272-1705. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Art Unit: 2814

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

Long Pham
Primary Examiner
Art Unit 2814

/LONG PHAM/
Primary Examiner, Art Unit 2814

<i>Index of Claims</i> 	Application/Control No. 15443276	Applicant(s)/Patent Under Reexamination STAMPER ET AL.
	Examiner LONG PHAM	Art Unit 2814

✓	Rejected	-	Cancelled	N	Non-Elected	A	Appeal
=	Allowed	÷	Restricted	I	Interference	O	Objected

☐ Claims renumbered in the same order as presented by applicant		☐ CPA		☐ T.D.		☐ R.1.47			
CLAIM		DATE							
Final	Original	07/18/2017							
	1	÷							
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	3	÷							
	4	÷							
	5	÷							
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	20	÷							

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Inventor(s) : Anthony K. STAMPER et al.

Group Art Unit: 2814

Appln. No. : 15/443,276

Examiner: PHAM, LONG

Filed : February 27, 2017

Confirmation No.: 2522

For : **SEGMENTED GUARD-RING AND CHIP EDGE SEALS**

RESPONSE TO RESTRICTION REQUIREMENT

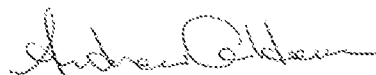
U.S. Patent and Trademark Office
Customer Service Window, Mail Stop _____
Randolph Building
401 Dulany Street
Alexandria, VA 22314

Sir:

In response to the Examiner's Restriction Requirement of July 24, 2017, the time set for response extending until September 25, 2017 (September 24, 2017 is being Sunday), Applicant hereby elects without traverse Group II, comprising Claims 1-10 and 12-19. Applicant further notes that claim 11 should also be included in this group as it has its dependency originating from claim 1. Claim 1 is generic.

If extensions of time are necessary to prevent abandonment of this application, then such extensions of time, associated exclusively with this response, are hereby petitioned under 37 C.F.R. §1.136(a), and any fees required therefore (including fees for net addition of claims) are hereby authorized to be charged to Deposit Account No. 50-2478.

Respectfully submitted,



Andrew M. Calderon
Reg. No. 38,093

Roberts, Mlotkowski, Safran, Cole & Calderon PC
P.O. Box 10064
McLean, VA 22102
Phone: 703.677.3011
Fax: 703.848.2981
Email: acalderon@rmsc2.com

Electronic Acknowledgement Receipt

EFS ID:	30174589
Application Number:	15443276
International Application Number:	
Confirmation Number:	2522
Title of Invention:	SEGMENTED GUARD-RING AND CHIP EDGE SEALS
First Named Inventor/Applicant Name:	Anthony K. STAMPER
Customer Number:	133809
Filer:	Andrew Michael Calderon/Aleksandra Paredes
Filer Authorized By:	Andrew Michael Calderon
Attorney Docket Number:	IBUR032-US-NP
Receipt Date:	24-AUG-2017
Filing Date:	27-FEB-2017
Time Stamp:	16:04:48
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
------------------------	----

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Response to Election / Restriction Filed	IBUR032-US-NP_Response to Restriction Requirement_08-24-17.pdf	25187 5db1266d0e908cda5f2bbbbae969695c69f54942c	no	1

Warnings:

Information:	
Total Files Size (in bytes):	25187
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.	



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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
15/443,276	02/27/2017	Anthony K. STAMPER	IBUR032-US-NP	2522
133809	7590	11/30/2017	EXAMINER	
Roberts Mlotkowski Safran Cole & Calderon, P.C.			PHAM, LONG	
Intellectual Property Department			ART UNIT	
P.O. Box 10064			PAPER NUMBER	
McLean, VA 22102-8064			2814	
			NOTIFICATION DATE	DELIVERY MODE
			11/30/2017	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

docketing@rmc2.com
lgalllaugh@rmc2.com
dbeltran@rmc2.com

Office Action Summary	Application No. 15/443,276	Applicant(s) STAMPER ET AL.	
	Examiner LONG PHAM	Art Unit 2814	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

- 1) ☒ Responsive to communication(s) filed on 8/24/17.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on ____.
- 2a) ☐ This action is **FINAL**. 2b) ☒ This action is non-final.
- 3) ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on ____; the restriction requirement and election have been incorporated into this action.
- 4) ☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

- 5) ☒ Claim(s) 1-20 is/are pending in the application.
5a) Of the above claim(s) 11 and 20 is/are withdrawn from consideration.
- 6) ☒ Claim(s) 12-19 is/are allowed.
- 7) ☒ Claim(s) 1,2 and 4 is/are rejected.
- 8) ☒ Claim(s) 3 and 5-10 is/are objected to.
- 9) ☐ Claim(s) ____ are subject to restriction and/or election requirement.

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

Application Papers

- 10) ☐ The specification is objected to by the Examiner.
- 11) ☒ The drawing(s) filed on 2/27/17 is/are: a) ☒ accepted or b) ☐ objected to by the Examiner.
Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).
Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

- 12) ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

- a) ☐ All b) ☐ Some** c) ☐ None of the:
1. ☐ Certified copies of the priority documents have been received.
 2. ☐ Certified copies of the priority documents have been received in Application No. ____.
 3. ☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

- | | |
|---|---|
| 1) ☒ Notice of References Cited (PTO-892) | 3) ☐ Interview Summary (PTO-413)
Paper No(s)/Mail Date. ____. |
| 2) ☐ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b)
Paper No(s)/Mail Date ____. | 4) ☐ Other: ____. |

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

DETAILED ACTION

Election/Restrictions

Applicant's election without traverse of claims 1-10 and 12-19 in the reply filed on 8/24/17 is acknowledged.

Claim Rejections - 35 USC § 103

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

Claims 1, 2, and 4 is/are rejected under 35 U.S.C. 103 as being unpatentable over the applicant's admitted prior art (AAPA) in combination with Tsutsue et al. (US pub 20120280401).

With respect to claim 1, AAPA teaches a structure comprising (see the Background on pages 1 and 2 of the specification of this application):

a guard ring structure formed in a low-k dielectric material formed to at least a substrate underneath the low-k dielectric material.

AAPA fails to teach the forming an edge seal structure formed through the low-k dielectric material to at least a substrate.

Tsutsue et al. teach a structure in which an edge seal structure 221a1, 221a2 formed through a dielectric material 205 to at least a substrate 201 underneath the dielectric material. See figs. 8A and 8B and associated text.

It would have been obvious to one of ordinary skill in the art of making semiconductor devices to incorporate the teaching of Tsutsue et al. into the device of AAPA to prevent the contaminant from entering the device. See para 1077.

With respect to claim 2, AAPA in combination with Tsutsue et al. teach the low-k dielectric material is low-K SiCOH or p-SiCOH middle of the line and/or back end of the line dielectric material.

With respect to claim 4, AAPA in combination with Tsutsue et al. teach the guard ring structure is a segmented guard ring structure.

Allowable Subject Matter

Claims 12-19 are allowed.

Claims 3, 5-10 are objected to as being dependent upon a rejected base claim, but would be allowable if rewritten in independent form including all of the limitations of the base claim and any intervening claims.

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to LONG PHAM whose telephone number is (571)272-1714. The examiner can normally be reached on Mon-Frid, 10am to 5pm.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at <http://www.uspto.gov/interviewpractice>.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Wael Fahmy can be reached on 571-272-1705. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

Long Pham
Primary Examiner
Art Unit 2814

Application/Control Number: 15/443,276
Art Unit: 2814

Page 5

/LONG PHAM/
Primary Examiner, Art Unit 2814

Notice of References Cited	Application/Control No. 15/443,276		Applicant(s)/Patent Under Reexamination STAMPER ET AL.	
	Examiner LONG PHAM		Art Unit 2814	Page 1 of 1

U.S. PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Name	CPC Classification	US Classification
*	A	US-2012/0280401 A1	11-2012	TSUTSUE; Makoto	H01L23/564	257/774
	B	US-				
	C	US-				
	D	US-				
	E	US-				
	F	US-				
	G	US-				
	H	US-				
	I	US-				
	J	US-				
	K	US-				
	L	US-				
	M	US-				

FOREIGN PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Country	Name	CPC Classification
	N					
	O					
	P					
	Q					
	R					
	S					
	T					

NON-PATENT DOCUMENTS

*		Include as applicable: Author, Title Date, Publisher, Edition or Volume, Pertinent Pages)
	U	
	V	
	W	
	X	

*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.05(a).)
Dates in MM-YYYY format are publication dates. Classifications may be US or foreign.

<i>Index of Claims</i> 	Application/Control No. 15443276	Applicant(s)/Patent Under Reexamination STAMPER ET AL.
	Examiner LONG PHAM	Art Unit 2814

✓	Rejected	-	Cancelled	N	Non-Elected	A	Appeal
=	Allowed	÷	Restricted	I	Interference	O	Objected

☐ Claims renumbered in the same order as presented by applicant				☐ CPA		☐ T.D.		☐ R.1.47	
CLAIM		DATE							
Final	Original	07/18/2017	11/24/2017						
	1	÷	✓						
	2	÷	✓						
	3	÷	O						
	4	÷	✓						
	5	÷	O						
	6	÷	O						
	7	÷	O						
	8	÷	O						
	9	÷	O						
	10	÷	O						
	11	÷	N						
	12	÷	=						
	13	÷	=						
	14	÷	=						
	15	÷	=						
	16	÷	=						
	17	÷	=						
	18	÷	=						
	19	÷	=						
	20	÷	N						

EAST Search History

EAST Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L3	477052	(seal or sealing or guard or ring) near3 (trench or trenches or grooves or groove)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/24 09:22
L4	372	3 near5 oxide	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/24 09:22
L5	10	4 near10 (air or gap)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/24 09:22
L6	126	4 same substrate	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/24 09:25
S4	2996	(seal or sealing) near4 (trench or trenches)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/10 07:45
S5	3748	(seal or sealing) near6 (trench or trenches)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/10 07:46
S6	305	S5 near8 (edge or peripheral or periphery)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/10 07:46
S7	734	S5 same (polyimide or polymer or oxide or (airgap or (air adj gap)))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/10 07:48
S8	46	S6 same8 S7	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/10 07:48
S9	2	(guard adj2 ring) same8 S8	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/10 07:48
S10	9	(guard adj2 ring) same8 S6	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/10 07:51
S11	7	S10 not S9	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/10 07:51

S12	575730	(sealing or seal) near8 (edge or peripheral or periphery)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/10 07:58
S13	490	(guard adj2 ring) same8 S12	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/10 07:58
S14	14	segmented and S13	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/10 07:59
S15	17	segmented adj2 guard adj2 ring	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/10 08:00
S16	26919	(guard adj2 ring)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/10 08:02
S17	31732	(low adj k adj dielectric) or (low-k adj dielectric)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/10 08:02
S18	183	S16 same8 S17	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/10 08:03
S19	604897	(seal or sealing or trench or trenches) near8 (edge or peripheral or periphery)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/10 08:04
S20	33	S18 same8 S19	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/10 08:04
S21	2	((("7517736") or ("8541262"))).PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2017/11/10 08:07
S22	576642	(sealing or seal) near8 (edge or peripheral or periphery)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/22 07:22
S23	490	(guard adj2 ring) same8 S22	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/22 07:22
S24	2	((("20150097257") or ("20150291415"))).PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2017/11/23 10:39
S25	302325	(seal or sealing) near6 (trench or trenches or groove or grooves)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/23 10:46
S26	496	S25 near10 dielectric	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/23 10:46
S27	20287	(seal or sealing) near6 (oxide or polyimide)	US-PGPUB; USPAT; USOCR; FPRS; EPO;	OR	ON	2017/11/23 10:47

			JPO; DERWENT; IBM_TDB			
S28	26	S26 same4 S27	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/23 10:47
S29	138924	(isolating or isolation) near6 (trench or trenches or groove or grooves)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/23 10:59
S30	69356	(isolating or isolation) near6 (oxide or polyimide)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/23 10:59
S31	14565	S29 near6 dielectric	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/23 11:00
S32	47970	S29 near6 substrate	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/23 11:00
S33	4093	S31 same S32	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/23 11:01
S34	566	S30 same S33	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/23 11:01
S35	1	(guard adj2 ring) same8 S34	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/23 11:01
S36	151	S34 same8 soi	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/23 11:03
S37	54	(seal or sealing) adj3 (through) adj3 dielectric	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/23 11:04
S38	6	S37 same substrate	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/23 11:05
S39	9591	(seal or sealing) near3 (oxide or polyimide)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/23 11:17
S40	1637	S39 same substrate	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/23 11:18
S41	356	S39 near4 substrate	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/23 11:18

S42	11	S39 near4 (through near4 dielectric)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/23 11:19
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11/ 24/ 2017 9:38:10 AM
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IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Inventor(s) : Anthony K. STAMPER et al. Group Art Unit: 2814
Appln. No. : 15/443,276 Examiner: PHAM, LONG
Filed : February 27, 2017 Confirmation No.: 2522
For : **SEGMENTED GUARD-RING AND CHIP EDGE SEALS**

AMENDMENT UNDER 37 C.F.R. §1.111

United States Patent and Trademark Office
Customer Service Window, Mail Stop Amendment
Randolph Building
401 Dulany Street
Alexandria, VA 22314

Sir:

In response to the Office Action dated November 30, 2017, please amend the above-identified application as follows.

A listing of claims begins on page 2.

Remarks begin on page 5.

If extensions of time are necessary to prevent abandonment of this application, then such extensions of time are hereby petitioned under 37 C.F.R. §1.136(a), and any fees required therefor (including fees for net addition of claims) are hereby authorized to be charged to our Deposit Account No. 50-2478.

AMENDMENT TO THE CLAIMS

Please **AMEND** claims 1, 4 and 10 as follows.

Please **CANCEL** claims 3 and 20 without prejudice or disclaimer.

A copy of all pending claims and a status of the claims are provided below.

1. (currently amended) A structure comprising:
a guard ring structure formed in a low-k dielectric material; and
an edge seal structure formed through the low-k dielectric material to at least a substrate underneath the low-k dielectric material,
wherein the substrate is a silicon on insulator substrate and the edge seal structure terminates below the low-k dielectric material.
2. (original) The structure of claim 1, wherein the low-k dielectric material is low-K SiCOH or p-SiCOH middle of the line and/or back end of the line dielectric material.
3. (canceled)
4. (currently amended) The structure of claim ~~[[3]]~~ 1, wherein the guard ring structure is a segmented guard ring structure.
5. (original) The structure of claim 4, wherein the edge seal structure is a trench lined with a passivation layer and filled with a polyimide.
6. (original) The structure of claim 5, wherein the polyimide is planarized above a last metal layer or bond pad.
7. (original) The structure of claim 1, wherein the edge seal structure is a trench filled with oxide material which is also covering a metal insulator metal structure.
8. (original) The structure of claim 7, wherein the trench filled with the oxide is below a last metal layer or bond pad.

9. (original) The structure of claim 8, wherein the trench filled with the oxide includes an airgap.
10. (currently amended) ~~The structure of claim 1~~ A structure comprising:
a guard ring structure formed in a low-k dielectric material; and
an edge seal structure formed through the low-k dielectric material to at least a substrate
underneath the low-k dielectric material,
wherein the edge seal structure is one or more airgaps formed in the low-k dielectric material that is pinched off with oxide.
11. (original) The structure of claim 10, wherein the oxide is a PECVD silane oxide.
12. (original) A structure comprising:
a silicon on insulator substrate;
a middle of the line and back end of the line dielectric low-k dielectric material on the silicon on insulator substrate;
a wiring structure formed in the middle of the line and back end of the line dielectric low-k dielectric material to the silicon on insulator substrate;
a segmented guard ring structure formed in the middle of the line and back end of the line dielectric low-k dielectric material; and
an edge seal structure formed extending through the middle of the line and back end of the line dielectric low-k dielectric material.
13. (original) The structure of claim 12, wherein the edge seal structure is a trench lined with a passivation layer and filled with a polyimide which has a planarized surface above a last metal layer or bond pad.
14. (original) The structure of claim 13, wherein the edge seal structure terminates below the middle of the line and back end of the line dielectric low-k dielectric material

15. (original) The structure of claim 12, wherein the edge seal structure is a trench filled with oxide material which covers a metal insulator metal structure below a last metal layer or bond pad.

16. (original) The structure of claim 15, wherein the trench filled with the oxide includes an airgap.

17. (original) The structure of claim 12, wherein the edge seal structure is one or more airgaps formed in the low-k dielectric material.

18. (original) The structure of claim 17, wherein the one or more airgaps are at a lower level of the wiring structure.

19. (original) The structure of claim 17, wherein the one or more airgaps are pinched off with PECVD silane oxide.

20. (canceled)

REMARKS

By this amendment, claims 1, 4 and 10 are amended for the Examiner's consideration. Claims 3 and 20 are canceled without prejudice or disclaimer. The above amendments do not add new matter to the application and are fully supported by the original claims and specification. Reconsideration of the rejected claims in view of the above amendments and the following remarks is respectfully requested.

Amendments to the Claims

In this Amendment, Applicant has amended claims and cancelled claims from further consideration in this application to facilitate expeditious prosecution of the application. Applicant is not conceding that the subject matter encompassed by the claims prior to this Amendment is unpatentable over the art cited by the Examiner. Applicant respectfully reserves the right to pursue claims in one or more continuing applications, including claims capturing the subject matter encompassed by such claims prior to this Amendment and additional claims.

Allowed Claims

Applicant appreciates the indication that claims 3 and 5-10 contain allowable subject matter. Applicant further appreciates the indication that claims 12-19 are allowed.

The allowable subject matter of claim 3 is now placed into independent claim 1. Accordingly, independent claim 1 and its dependent claims are now in condition for allowance. In addition, allowable claim 10 has also been revised into independent format to include the features of base claim 1. Accordingly, claim 10 and its dependent claim 11 are also in condition for allowance. With these amendments, Applicant submits that all of the claims are in condition for allowance.

35 U.S.C. §103 Rejection

Claims 1, 2, and 4 are rejected under 35 U.S.C. §103 for being unpatentable over applicant's admitted prior art (AAPA) in combination with U.S. Pub. No. 20120280401 to Tsutsue. This rejection is respectfully traversed.

Applicant first notes that the rejection of claim 4 is improper. Claim 4 is a dependent claim, depending on allowable claim 3. Accordingly, claim 4 by virtue of this dependency should also be considered allowable.

Applicant further submits that the rejection of claims 1 and 2 (and claim 4) are moot in view of the above amendments. Accordingly, Applicant respectfully requests that the rejection over claims 1, 2 and 4 be withdrawn.

Applicant also submits that the specification does not provide any AAPA, much less that a guard ring structure formed in a low-k dielectric material formed to at least a substrate underneath the low-k dielectric material. In fact, Applicant submits that the background section of the specification does not even mention such specific features.

CONCLUSION

In view of the foregoing, Applicant submits that all of the claims are patentably distinct from the prior art of record and are in condition for allowance. The Examiner is respectfully requested to pass the above application to issue. Further, any amendments to the claims which have been made in this response and which have not been specifically noted to overcome a rejection based upon the prior art, should be considered to have been made for a purpose unrelated to patentability, and no estoppel should be deemed to attach thereto. The Examiner is invited to contact the undersigned at the telephone number listed below, if needed. Applicant hereby makes a written conditional petition for extension of time, if required. Please charge any deficiencies in fees and credit any overpayment of fees to Deposit Account No. 50-2478.

Respectfully submitted,



Andrew M. Calderon
Reg. No. 38,093

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P.O. Box 10064
McLean, VA 22102
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Fax: 703.848.2981
Email: acalderon@rmcsc2.com

Electronic Acknowledgement Receipt

EFS ID:	31913894
Application Number:	15443276
International Application Number:	
Confirmation Number:	2522
Title of Invention:	SEGMENTED GUARD-RING AND CHIP EDGE SEALS
First Named Inventor/Applicant Name:	Anthony K. STAMPER
Customer Number:	133809
Filer:	Andrew Michael Calderon/Aleksandra Paredes
Filer Authorized By:	Andrew Michael Calderon
Attorney Docket Number:	IBUR032-US-NP
Receipt Date:	28-FEB-2018
Filing Date:	27-FEB-2017
Time Stamp:	14:58:20
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Amendment/Req. Reconsideration-After Non-Final Reject	IBUR032-US-NP_AmendmenttoOfficeAction_02-28-18.pdf	207533 48fa3ea97d951d2a8c64d134791a9b020c87dc95	no	6

Warnings:

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Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875	Application or Docket Number 15/443,276	Filing Date 02/27/2017	☐ To be Mailed
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ENTITY: ☒ LARGE ☐ SMALL ☐ MICRO**APPLICATION AS FILED – PART I**

(Column 1)

(Column 2)

FOR	NUMBER FILED	NUMBER EXTRA	RATE (\$)	FEE (\$)
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A	
☐ SEARCH FEE (37 CFR 1.16(k), (i), or (m))	N/A	N/A	N/A	
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A	
TOTAL CLAIMS (37 CFR 1.16(i))	minus 20 =	*	X \$ =	
INDEPENDENT CLAIMS (37 CFR 1.16(h))	minus 3 =	*	X \$ =	
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).			
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))				
			TOTAL	

* If the difference in column 1 is less than zero, enter "0" in column 2.

APPLICATION AS AMENDED – PART II

(Column 1)

(Column 2)

(Column 3)

AMENDMENT	02/28/2018	CLAIMS REMAINING AFTER AMENDMENT	MINUS	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)
	Total (37 CFR 1.16(i))	* 17	Minus	** 20	=	X \$ =	
	Independent (37 CFR 1.16(h))	* 3	Minus	*** 3	=	X \$ =	
	☐ Application Size Fee (37 CFR 1.16(s))						
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						
						TOTAL ADD'L FEE	

(Column 1)

(Column 2)

(Column 3)

AMENDMENT	CLAIMS REMAINING AFTER AMENDMENT	MINUS	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)
	Total (37 CFR 1.16(i))	*	Minus	**	=	X \$ =
	Independent (37 CFR 1.16(h))	*	Minus	***	=	X \$ =
	☐ Application Size Fee (37 CFR 1.16(s))					
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))					
					TOTAL ADD'L FEE	

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".

*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".

The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.

LIE
TERRANCE LAWRENCE

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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NOTICE OF ALLOWANCE AND FEE(S) DUE

133809 7590 05/11/2018
Roberts Mlotkowski Safran Cole & Calderon, P.C.
Intellectual Property Department
P.O. Box 10064
McLean, VA 22102-8064

EXAMINER

PHAM, LONG

ART UNIT

PAPER NUMBER

2814

DATE MAILED: 05/11/2018

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
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15/443,276

02/27/2017

Anthony K. STAMPER

IBUR032-US-NP

2522

TITLE OF INVENTION: SEGMENTED GUARD-RING AND CHIP EDGE SEALS

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1000	\$0	\$0	\$1000	08/13/2018

THE APPLICATION IDENTIFIED ABOVE HAS BEEN EXAMINED AND IS ALLOWED FOR ISSUANCE AS A PATENT. PROSECUTION ON THE MERITS IS CLOSED. THIS NOTICE OF ALLOWANCE IS NOT A GRANT OF PATENT RIGHTS. THIS APPLICATION IS SUBJECT TO WITHDRAWAL FROM ISSUE AT THE INITIATIVE OF THE OFFICE OR UPON PETITION BY THE APPLICANT. SEE 37 CFR 1.313 AND MPEP 1308.

THE ISSUE FEE AND PUBLICATION FEE (IF REQUIRED) MUST BE PAID WITHIN THREE MONTHS FROM THE MAILING DATE OF THIS NOTICE OR THIS APPLICATION SHALL BE REGARDED AS ABANDONED. THIS STATUTORY PERIOD CANNOT BE EXTENDED. SEE 35 U.S.C. 151. THE ISSUE FEE DUE INDICATED ABOVE DOES NOT REFLECT A CREDIT FOR ANY PREVIOUSLY PAID ISSUE FEE IN THIS APPLICATION. IF AN ISSUE FEE HAS PREVIOUSLY BEEN PAID IN THIS APPLICATION (AS SHOWN ABOVE), THE RETURN OF PART B OF THIS FORM WILL BE CONSIDERED A REQUEST TO REAPPLY THE PREVIOUSLY PAID ISSUE FEE TOWARD THE ISSUE FEE NOW DUE.

HOW TO REPLY TO THIS NOTICE:

I. Review the ENTITY STATUS shown above. If the ENTITY STATUS is shown as SMALL or MICRO, verify whether entitlement to that entity status still applies.

If the ENTITY STATUS is the same as shown above, pay the TOTAL FEE(S) DUE shown above.

If the ENTITY STATUS is changed from that shown above, on PART B - FEE(S) TRANSMITTAL, complete section number 5 titled "Change in Entity Status (from status indicated above)".

For purposes of this notice, small entity fees are 1/2 the amount of undiscounted fees, and micro entity fees are 1/2 the amount of small entity fees.

II. PART B - FEE(S) TRANSMITTAL, or its equivalent, must be completed and returned to the United States Patent and Trademark Office (USPTO) with your ISSUE FEE and PUBLICATION FEE (if required). If you are charging the fee(s) to your deposit account, section "4b" of Part B - Fee(s) Transmittal should be completed and an extra copy of the form should be submitted. If an equivalent of Part B is filed, a request to reapply a previously paid issue fee must be clearly made, and delays in processing may occur due to the difficulty in recognizing the paper as an equivalent of Part B.

III. All communications regarding this application must give the application number. Please direct all communications prior to issuance to Mail Stop ISSUE FEE unless advised to the contrary.

IMPORTANT REMINDER: Maintenance fees are due in utility patents issuing on applications filed on or after Dec. 12, 1980. It is patentee's responsibility to ensure timely payment of maintenance fees when due. More information is available at www.uspto.gov/PatentMaintenanceFees.

PART B - FEE(S) TRANSMITTAL

**Complete and send this form, together with applicable fee(s), to: Mail Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450
or Fax (571)-273-2885**

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence including the Patent, advance orders and notification of maintenance fees will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications.

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

133809 7590 05/11/2018
Roberts Mlotkowski Safran Cole & Calderon, P.C.
Intellectual Property Department
P.O. Box 10064
McLean, VA 22102-8064

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being facsimile transmitted to the USPTO (571) 273-2885, on the date indicated below.

(Depositor's name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
15/443,276	02/27/2017	Anthony K. STAMPER	IBUR032-US-NP	2522

TITLE OF INVENTION: SEGMENTED GUARD-RING AND CHIP EDGE SEALS

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1000	\$0	\$0	\$1000	08/13/2018

EXAMINER	ART UNIT	CLASS-SUBCLASS
PHAM, LONG	2814	257-491000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363). ☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached. ☐ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47; Rev 03-02 or more recent) attached. Use of a Customer Number is required.	2. For printing on the patent front page, list (1) The names of up to 3 registered patent attorneys or agents OR, alternatively, 1 _____ (2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed. 2 _____ 3 _____
---	---

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document has been filed for recordation as set forth in 37 CFR 3.11. Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE (B) RESIDENCE: (CITY and STATE OR COUNTRY)

Please check the appropriate assignee category or categories (will not be printed on the patent) : ☐ Individual ☐ Corporation or other private group entity ☐ Government

4a. The following fee(s) are submitted:

- ☐ Issue Fee
☐ Publication Fee (No small entity discount permitted)
☐ Advance Order - # of Copies _____

4b. Payment of Fee(s): (Please first reapply any previously paid issue fee shown above)

- ☐ A check is enclosed.
☐ Payment by credit card. Form PTO-2038 is attached.
☐ The director is hereby authorized to charge the required fee(s), any deficiency, or credits any overpayment, to Deposit Account Number _____ (enclose an extra copy of this form).

5. Change in Entity Status (from status indicated above)

- ☐ Applicant certifying micro entity status. See 37 CFR 1.29
☐ Applicant asserting small entity status. See 37 CFR 1.27
☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature _____ Date _____
Typed or printed name _____ Registration No. _____



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
15/443,276	02/27/2017	Anthony K. STAMPER	IBUR032-US-NP	2522

133809 7590 05/11/2018
Roberts Mlotkowski Safran Cole & Calderon, P.C.
Intellectual Property Department
P.O. Box 10064
McLean, VA 22102-8064

EXAMINER

PHAM, LONG

ART UNIT	PAPER NUMBER
----------	--------------

2814

DATE MAILED: 05/11/2018

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (Applications filed on or after May 29, 2000)

The Office has discontinued providing a Patent Term Adjustment (PTA) calculation with the Notice of Allowance.

Section 1(h)(2) of the AIA Technical Corrections Act amended 35 U.S.C. 154(b)(3)(B)(i) to eliminate the requirement that the Office provide a patent term adjustment determination with the notice of allowance. See Revisions to Patent Term Adjustment, 78 Fed. Reg. 19416, 19417 (Apr. 1, 2013). Therefore, the Office is no longer providing an initial patent term adjustment determination with the notice of allowance. The Office will continue to provide a patent term adjustment determination with the Issue Notification Letter that is mailed to applicant approximately three weeks prior to the issue date of the patent, and will include the patent term adjustment on the patent. Any request for reconsideration of the patent term adjustment determination (or reinstatement of patent term adjustment) should follow the process outlined in 37 CFR 1.705.

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Customer Service Center of the Office of Patent Publication at 1-(888)-786-0101 or (571)-272-4200.

OMB Clearance and PRA Burden Statement for PTOL-85 Part B

The Paperwork Reduction Act (PRA) of 1995 requires Federal agencies to obtain Office of Management and Budget approval before requesting most types of information from the public. When OMB approves an agency request to collect information from the public, OMB (i) provides a valid OMB Control Number and expiration date for the agency to display on the instrument that will be used to collect the information and (ii) requires the agency to inform the public about the OMB Control Number's legal significance in accordance with 5 CFR 1320.5(b).

The information collected by PTOL-85 Part B is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450. Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Notice of Allowability	Application No. 15/443,276	Applicant(s) STAMPER ET AL.	
	Examiner LONG PHAM	Art Unit 2814	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to AMENDMENT DATED 2/28/2018.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

3. ☒ The allowed claim(s) is/are 1, 2, 4-19. As a result of the allowed claim(s), you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

4. ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a) ☐ All b) ☐ Some *c) ☐ None of the:

1. ☐ Certified copies of the priority documents have been received.

2. ☐ Certified copies of the priority documents have been received in Application No. _____.

3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: _____.

Applicant has THREE MONTHS FROM THE "MAILING DATE" of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.
THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date _____.

Identifying indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).

6. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s) 1. ☐ Notice of References Cited (PTO-892) 2. ☐ Information Disclosure Statements (PTO/SB/08), Paper No./Mail Date _____ 3. ☐ Examiner's Comment Regarding Requirement for Deposit of Biological Material 4. ☐ Interview Summary (PTO-413), Paper No./Mail Date _____ 5. ☐ Examiner's Amendment/Comment 6. ☐ Examiner's Statement of Reasons for Allowance 7. ☐ Other _____.	
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/ LONG PHAM/ Primary Examiner, Art Unit 2814	
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BIB DATA SHEET

CONFIRMATION NO. 2522

SERIAL NUMBER	FILING or 371(c) DATE	CLASS	GROUP ART UNIT	ATTORNEY DOCKET NO.		
15/443,276	02/27/2017	257	2814	IBUR032-US-NP		
RULE						
APPLICANTS GLOBALFOUNDRIES INC., GRAND CAYMAN, CAYMAN ISLANDS; INVENTORS Anthony K. STAMPER, Burlington, VT; Vincent J. MCGAHAY, Poughkeepsie, NY; Zhong-Xiang HE, Essex Junction, VT; ** CONTINUING DATA ***** ** FOREIGN APPLICATIONS ***** ** IF REQUIRED, FOREIGN FILING LICENSE GRANTED ** 03/07/2017						
Foreign Priority claimed ☐ Yes ☒ No 35 USC 119(a-d) conditions met ☐ Yes ☒ No Verified and Acknowledged <u>/LONG PHAM/</u> Examiner's Signature		☐ Met after Allowance Initials	STATE OR COUNTRY VT	SHEETS DRAWINGS 7	TOTAL CLAIMS 20	INDEPENDENT CLAIMS 3
ADDRESS Roberts Mlotkowski Safran Cole & Calderon, P.C. Intellectual Property Department P.O. Box 10064 McLean, VA 22102-8064 UNITED STATES						
TITLE SEGMENTED GUARD-RING AND CHIP EDGE SEALS						
FILING FEE RECEIVED 1600	FEES: Authority has been given in Paper No. _____ to charge/credit DEPOSIT ACCOUNT No. _____ for following:			☐ All Fees ☐ 1.16 Fees (Filing) ☐ 1.17 Fees (Processing Ext. of time) ☐ 1.18 Fees (Issue) ☐ Other _____ ☐ Credit		

Search Notes 	Application/Control No. 15443276	Applicant(s)/Patent Under Reexamination STAMPER ET AL.
	Examiner LONG PHAM	Art Unit 2814

CPC- SEARCHED		
Symbol	Date	Examiner
H01L29/0619,H01L23/3171,H01L29/0649,H01L23/528	4/25/2018	LP

CPC COMBINATION SETS - SEARCHED		
Symbol	Date	Examiner

US CLASSIFICATION SEARCHED			
Class	Subclass	Date	Examiner

* See search history printout included with this form or the SEARCH NOTES box below to determine the scope of the search.

SEARCH NOTES		
Search Notes	Date	Examiner
UPDATED TEXT AND INTERFERENCE SEARCHES	4/18/2018	LP

INTERFERENCE SEARCH			
US Class/ CPC Symbol	US Subclass / CPC Group	Date	Examiner
H01L	29/0619	4/25/2018	LP

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<i>Index of Claims</i> 	Application/Control No. 15443276	Applicant(s)/Patent Under Reexamination STAMPER ET AL.
	Examiner LONG PHAM	Art Unit 2814

✓	Rejected	-	Cancelled	N	Non-Elected	A	Appeal
=	Allowed	÷	Restricted	I	Interference	O	Objected

☐ Claims renumbered in the same order as presented by applicant					☐ CPA					☐ T.D.					☐ R.1.47				
CLAIM		DATE																	
Final	Original	07/18/2017	11/24/2017	04/25/2018															
	1	÷	✓	=															
	2	÷	✓	=															
	3	÷	O	-															
	4	÷	✓	=															
	5	÷	O	=															
	6	÷	O	=															
	7	÷	O	=															
	8	÷	O	=															
	9	÷	O	=															
	10	÷	O	=															
	11	÷	N	=															
	12	÷	=	=															
	13	÷	=	=															
	14	÷	=	=															
	15	÷	=	=															
	16	÷	=	=															
	17	÷	=	=															
	18	÷	=	=															
	19	÷	=	=															
	20	÷	N	-															

EAST Search History

EAST Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L21	12073	H01L29/0619.CPC.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/04/25 10:32
L23	7410	H01L23/3171.CPC.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/04/25 10:32
L24	14597	H01L29/0649.CPC.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/04/25 10:32
L25	14885	H01L23/528.CPC.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/04/25 10:32
S4	2996	(seal or sealing) near4 (trench or trenches)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/10 07:45
S5	3748	(seal or sealing) near6 (trench or trenches)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/10 07:46
S6	305	S5 near8 (edge or peripheral or periphery)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/10 07:46
S7	734	S5 same (polyimide or polymer or oxide or (airgap or (air adj gap)))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/10 07:48
S8	46	S6 same8 S7	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/10 07:48
S9	2	(guard adj2 ring) same8 S8	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/10 07:48
S10	9	(guard adj2 ring) same8 S6	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/10 07:51
S11	7	S10 not S9	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/10 07:51

S12	575730	(sealing or seal) near8 (edge or peripheral or periphery)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/10 07:58
S13	490	(guard adj2 ring) same8 S12	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/10 07:58
S14	14	segmented and S13	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/10 07:59
S15	17	segmented adj2 guard adj2 ring	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/10 08:00
S16	26919	(guard adj2 ring)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/10 08:02
S17	31732	(low adj k adj dielectric) or (low-k adj dielectric)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/10 08:02
S18	183	S16 same8 S17	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/10 08:03
S19	604897	(seal or sealing or trench or trenches) near8 (edge or peripheral or periphery)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/10 08:04
S20	33	S18 same8 S19	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/10 08:04
S21	2	((("7517736") or ("8541262"))).PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2017/11/10 08:07
S22	576642	(sealing or seal) near8 (edge or peripheral or periphery)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/22 07:22
S23	490	(guard adj2 ring) same8 S22	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/22 07:22
S24	2	((("20150097257") or ("20150291415"))).PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2017/11/23 10:39
S25	302325	(seal or sealing) near6 (trench or trenches or groove or grooves)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/23 10:46
S26	496	S25 near10 dielectric	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/23 10:46
S27	20287	(seal or sealing) near6 (oxide or polyimide)	US-PGPUB; USPAT; USOCR; FPRS; EPO;	OR	ON	2017/11/23 10:47

			JPO; DERWENT; IBM_TDB			
S28	26	S26 same4 S27	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/23 10:47
S29	138924	(isolating or isolation) near6 (trench or trenches or groove or grooves)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/23 10:59
S30	69356	(isolating or isolation) near6 (oxide or polyimide)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/23 10:59
S31	14565	S29 near6 dielectric	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/23 11:00
S32	47970	S29 near6 substrate	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/23 11:00
S33	4093	S31 same S32	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/23 11:01
S34	566	S30 same S33	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/23 11:01
S35	1	(guard adj2 ring) same8 S34	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/23 11:01
S36	151	S34 same8 soi	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/23 11:03
S37	54	(seal or sealing) adj3 (through) adj3 dielectric	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/23 11:04
S38	6	S37 same substrate	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/23 11:05
S39	9591	(seal or sealing) near3 (oxide or polyimide)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/23 11:17
S40	1637	S39 same substrate	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/23 11:18
S41	356	S39 near4 substrate	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/23 11:18

S42	11	S39 near4 (through near4 dielectric)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/23 11:19
S43	477052	(seal or sealing or guard or ring) near3 (trench or trenches or grooves or groove)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/24 09:22
S44	372	S43 near5 oxide	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/24 09:22
S45	10	S44 near10 (air or gap)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/24 09:22
S46	126	S44 same substrate	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/24 09:25
S47	3759	(seal or sealing) near6 (trench or trenches)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/24 09:49
S48	305	S47 near8 (edge or peripheral or periphery)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/24 09:49
S49	3759	(seal or sealing) near6 (trench or trenches)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/24 10:24
S50	738	S49 same (polyimide or polymer or oxide or (airgap or (air adj gap)))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/11/24 10:24

4/ 25/ 2018 10:34:20 AM
C:\ Users\ lpham\ Documents\ EAST\ Workspaces\ 15-443,276.wsp

Issue Classification 	Application/Control No. 15443276	Applicant(s)/Patent Under Reexamination STAMPER ET AL.
	Examiner LONG PHAM	Art Unit 2814

☐ Claims renumbered in the same order as presented by applicant															
☐ CPA															
☐ T.D.															
☐ R.1.47															
Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original
1	1	17	18												
2	2	18	19												
3	4														
4	5														
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9	10														
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12	13														
13	14														
14	15														
15	16														
16	17														

NONE	Total Claims Allowed: 18	
(Assistant Examiner)	(Date)	
/LONG PHAM/ Primary Examiner.Art Unit 2814	04/25/2018	O.G. Print Claim(s) 1
(Primary Examiner)	(Date)	O.G. Print Figure 3

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

"FEE ADDRESS" INDICATION FORM

Address to:
Mail Stop M Correspondence
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Fax to:
571-273-6500

- OR -

INSTRUCTIONS: The issue fee must have been paid for application(s) listed on this form. In addition, only an address represented by a Customer Number can be established as the fee address for maintenance fee purposes (hereafter, fee address). A fee address should be established when correspondence related to maintenance fees should be mailed to a different address than the correspondence address for the application. **When to check the first box below:** If you have a Customer Number to represent the fee address. **When to check the second box below:** If you have no Customer Number representing the desired fee address, in which case a completed Request for Customer Number (PTO/SB/125) must be attached to this form. For more information on Customer Numbers, see the Manual of Patent Examining Procedure (MPEP) § 403.

For the following listed application(s), please recognize as the "Fee Address" under the provisions of 37 CFR 1.363 the address associated with:



Customer Number:

00197**OR**

The attached Request for Customer Number (PTO/SB/125) form.

PATENT NUMBER (if known)	APPLICATION NUMBER
	15/443,276

Completed by (check one):



Applicant/Inventor

/Andrew M. Calderon/

Signature



Attorney or Agent of record 38,093
 (Reg. No.)

Andrew M. Calderon

Typed or printed name



Assignee of record of the entire interest. See 37 CFR 3.71.
 Statement under 37 CFR 3.73(b) is enclosed.
 (Form PTO/SB/96)

703-677-3011

Requester's telephone number



Assignee recorded at Reel _____ Frame _____

July 11, 2018

Date

NOTE: Signatures of all the inventors or assignees of record of the entire interest or their representative(s) are required. Submit multiple forms if more than one signature is required, see below*.

* Total of 1 forms are submitted.

This collection of information is required by 37 CFR 1.363. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 5 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND COMPLETE D FORMS TO THIS ADDRESS.
SEND TO: Mail Stop M Correspondence, Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

Privacy Act Statement

The **Privacy Act of 1974 (P.L. 93-579)** requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (*i.e.*, GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Electronic Patent Application Fee Transmittal

Application Number:	15443276			
Filing Date:	27-Feb-2017			
Title of Invention:	SEGMENTED GUARD-RING AND CHIP EDGE SEALS			
First Named Inventor/Applicant Name:	Anthony K. STAMPER			
Filer:	Andrew Michael Calderon/Aleksandra Paredes			
Attorney Docket Number:	IBUR032-US-NP			
Filed as Large Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
PUBL. FEE- EARLY, VOLUNTARY, OR NORMAL	1504	1	0	0
UTILITY APPL ISSUE FEE	1501	1	1000	1000
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				1000

Electronic Acknowledgement Receipt

EFS ID:	33150524
Application Number:	15443276
International Application Number:	
Confirmation Number:	2522
Title of Invention:	SEGMENTED GUARD-RING AND CHIP EDGE SEALS
First Named Inventor/Applicant Name:	Anthony K. STAMPER
Customer Number:	133809
Filer:	Andrew Michael Calderon/Aleksandra Paredes
Filer Authorized By:	Andrew Michael Calderon
Attorney Docket Number:	IBUR032-US-NP
Receipt Date:	11-JUL-2018
Filing Date:	27-FEB-2017
Time Stamp:	16:13:17
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	CARD
Payment was successfully received in RAM	\$ 1000
RAM confirmation Number	071218INTEFSW16134100
Deposit Account	502478
Authorized User	Andrew Calderon

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:
37 CFR 1.20 (Post Issuance fees)

File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Issue Fee Payment (PTO-85B)	IBUR032-US- NP_IssueFeePayment_07-11-18 .pdf	91843	no	1
			55a5d80e681e361d194b58b532e5ffc727c71d48		
Warnings:					
Information:					
2	Change of Address	IBUR032-US- NP_ChangeofAddress_07-11-18.pdf	312616	no	2
			92e2fd1c98237074131f3aa752a655f981c0e49		
Warnings:					
Information:					
3	Fee Worksheet (SB06)	fee-info.pdf	31905	no	2
			3544dae7cce6d540052343909e887ac2cff53ff1		
Warnings:					
Information:					
Total Files Size (in bytes):			436364		
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					

PART B - FEE(S) TRANSMITTAL

Complete and send this form, together with applicable fee(s), to: **Mail** Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450
or Fax (571)-273-2885

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence including the Patent, advance orders and notification of maintenance fees will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications.

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being facsimile transmitted to the USPTO (571) 273-2885, on the date indicated below.

(Depositor's name)
(Signature)
(Date)

133809 7590 05/11/2018
Roberts Mlotkowski Safran Cole & Calderon, P.C.
Intellectual Property Department
P.O. Box 10064
McLean, VA 22102-8064

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
15/443,276	02/27/2017	Anthony K. STAMPER	IBUR032-US-NP	2522

TITLE OF INVENTION: SEGMENTED GUARD-RING AND CHIP EDGE SEALS

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1000	\$0	\$0	\$1000	08/13/2018

EXAMINER	ART UNIT	CLASS-SUBCLASS
PHAM, LONG	2814	257-491000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).
☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached.
☒ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47; Rev 03-02 or more recent) attached. Use of a Customer Number is required.

2. For printing on the patent front page, list
 (1) The names of up to 3 registered patent attorneys or agents OR, alternatively,
 (2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.

1 Anthony Canale
 2 Andrew M. Calderon
 3 Roberts Mlotkowski Safran
Cole & Calderon, P.C.

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document has been filed for recordation as set forth in 37 CFR 3.11. Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE

GLOBALFOUNDRIES INC.

(B) RESIDENCE: (CITY and STATE OR COUNTRY)

GRAND CAYMAN, CAYMAN ISLANDS

Please check the appropriate assignee category or categories (will not be printed on the patent): ☐ Individual ☒ Corporation or other private group entity ☐ Government

4a. The following fee(s) are submitted:

- ☒ Issue Fee
☒ Publication Fee (No small entity discount permitted)
☐ Advance Order - # of Copies _____

4b. Payment of Fee(s): (Please first reapply any previously paid issue fee shown above)

- ☐ A check is enclosed.
☒ Payment by credit card.
☒ The director is hereby authorized to charge the required fee(s), any deficiency, or credits any overpayment, to Deposit Account Number 50-2478 (enclose an extra copy of this form).

5. Change in Entity Status (from status indicated above)

- ☐ Applicant certifying micro entity status. See 37 CFR 1.29
☐ Applicant asserting small entity status. See 37 CFR 1.27
☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature Andrew M. Calderon
 Typed or printed name Andrew M. Calderon

Date 7/11/2018
 Registration No. 38,093



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United States Patent and Trademark Office
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www.uspto.gov

APPLICATION NO.	ISSUE DATE	PATENT NO.	ATTORNEY DOCKET NO.	CONFIRMATION NO.
15/443,276	08/28/2018	10062748	IBUR032-US-NP	2522

133809 7590 08/08/2018

Roberts Mlotkowski Safran Cole & Calderon, P.C.
Intellectual Property Department
P.O. Box 10064
McLean, VA 22102-8064

ISSUE NOTIFICATION

The projected patent number and issue date are specified above.

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (application filed on or after May 29, 2000)

The Patent Term Adjustment is 0 day(s). Any patent to issue from the above-identified application will include an indication of the adjustment on the front page.

If a Continued Prosecution Application (CPA) was filed in the above-identified application, the filing date that determines Patent Term Adjustment is the filing date of the most recent CPA.

Applicant will be able to obtain more detailed information by accessing the Patent Application Information Retrieval (PAIR) WEB site (<http://pair.uspto.gov>).

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Application Assistance Unit (AAU) of the Office of Data Management (ODM) at (571)-272-4200.

APPLICANT(s) (Please see PAIR WEB site <http://pair.uspto.gov> for additional applicants):

Anthony K. STAMPER, Burlington, VT;
GLOBALFOUNDRIES INC., GRAND CAYMAN, CAYMAN ISLANDS;
Vincent J. MCGAHAY, Poughkeepsie, NY;
Zhong-Xiang HE, Essex Junction, VT;

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UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
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Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
15/443,276	02/27/2017	Anthony K. STAMPER	IBUR032-US-NP

CONFIRMATION NO. 2522

PUBLICATION NOTICE



OC000000101984491

133809
Roberts Mlotkowski Safran Cole & Calderon, P.C.
Intellectual Property Department
P.O. Box 10064
McLean, VA 22102-8064

Title:SEGMENTED GUARD-RING AND CHIP EDGE SEALS

Publication No.US-2018-0248001-A1

Publication Date:08/30/2018

NOTICE OF PUBLICATION OF APPLICATION

The above-identified application will be electronically published as a patent application publication pursuant to 37 CFR 1.211, et seq. The patent application publication number and publication date are set forth above.

The publication may be accessed through the USPTO's publically available Searchable Databases via the Internet at www.uspto.gov. The direct link to access the publication is currently <http://www.uspto.gov/patft/>.

The publication process established by the Office does not provide for mailing a copy of the publication to applicant. A copy of the publication may be obtained from the Office upon payment of the appropriate fee set forth in 37 CFR 1.19(a)(1). Orders for copies of patent application publications are handled by the USPTO's Public Records Division. The Public Records Division can be reached by telephone at (571) 272-3150 or (800) 972-6382, by facsimile at (571) 273-3250, by mail addressed to the United States Patent and Trademark Office, Public Records Division, Alexandria, VA 22313-1450 or via the Internet.

In addition, information on the status of the application, including the mailing date of Office actions and the dates of receipt of correspondence filed in the Office, may also be accessed via the Internet through the Patent Electronic Business Center at www.uspto.gov using the public side of the Patent Application Information and Retrieval (PAIR) system. The direct link to access this status information is currently <https://portal.uspto.gov/pair/PublicPair>. Prior to publication, such status information is confidential and may only be obtained by applicant using the private side of PAIR.

Further assistance in electronically accessing the publication, or about PAIR, is available by calling the Patent Electronic Business Center at 1-866-217-9197.

Office of Data Management, Application Assistance Unit (571) 272-4000, or (571) 272-4200, or 1-888-786-0101